

Properties and applications of shape-stabilized phase change energy storage materials based on porous material support—A review

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ABSTRACT

Advanced phase change energy storage technology can solve the contradiction between time and space energy supply and demand and improve energy efficiency. It is considered one of the most effective strategies to utilize various renewable energy in energy saving and environmental protection. Solid-liquid phase change materials (PCMs) have become critical in developing thermal energy storage (TES) technology because of their high energy storage density, high latent heat, and excellent constant temperature performance during phase change. However, volume expansion, leakage, and low thermal conductivity during phase change are the main problems hindering the development of phase change materials. To overcome these inherent problems and improve their thermophysical properties, it has been identified as an effective way to construct composite phase change materials (ss-PCMs) with stable shapes by encapsulating PCMs with porous materials, which can effectively prevent the leakage of solid-liquid phase change and improve the heat transfer performance of ss-PCMs. This paper reviews the main research progress of porous support materials (such as metal foam, porous polymer, carbon-based three-dimensional porous materials, porous ceramic materials, etc.) as ss-PCMs supports to evaluate the advantages and disadvantages of porous materials. The effects of different porous materials on the thermophysical properties of PCMs, such as thermal conductivity, latent heat, phase transition temperature, undercooling, shape stability, thermal cycle stability, and chemical compatibility, were introduced in detail. In addition, the applications of different porous material-based composite phase change materials in various industries are summarized. Finally, the research topics and challenges of porous materials in the future development of ss-PCMs are summarized and prospected.

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1. Introduction

With the development of the economy in recent years, the demand for energy and the consumption of non-renewable resources has increased year by year. In addition, with the use of fossil fuels, greenhouse gas emissions are growing, and the global climate is seriously affected. The demand for developing clean and renewable energy is becoming more urgent. However, renewable energy

sources such as solar and wind are cyclical and cannot be collected at night or without wind. Energy storage technology can solve the contradiction between time and space energy supply and demand and improve energy utilization efficiency. Developing new energy storage technology is considered one of the most effective strategies for utilizing various renewable energy sources for energy saving and environmental protection. Up to now, energy storage technology mainly includes mechanical energy, electrical energy, chemical energy, and thermal energy storage. Mechanical energy is usually stored in the form of kinetic energy or potential energy [1]. Large mechanical energy storage mainly uses compressed air and pumped storage [2,3]. Electrical energy is primarily stored in batteries in the form of chemical energy [4]. Thermal energy storage is

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another important way, and its storage and release are realized by changing the internal energy of materials. Among all kinds of energy, thermal energy has attracted more and more attention and extensive research because of its widespread existence [5–8].

Generally speaking, heat storage technology can be divided into chemical heat storage technology and physical heat storage technology, among which physical heat storage technology can be divided into sensible heat storage technology and phase change (latent heat) heat storage technology [9]. Table 1 compares thermochemical, sensible heat, and phase change (latent heat) heat storage technologies. Chemical heat storage technology is to store and releases heat through a combination of reversible chemical reactions. The chemical heat storage system has the advantage of high energy storage density, but the disadvantage of this heat storage method is that the system is very complex, and the cost is too high. Therefore, it is still in the research stage in the laboratory [10]; Sensible heat storage technology uses the change of material temperature difference to realize heat storage. Its principle is simple, the cost is low, and the technology maturity is high. However, the density of material energy storage is relatively low, the volume of equipment is relatively large, the stored heat energy cannot be released at a certain temperature when releasing heat energy, and its temperature change is continuous [11,12]; Phase change (latent heat) heat storage technology is to store and release heat by using the change of latent heat of phase change materials (PCMs) during phase change. This technology has the advantages of high heat storage density, constant temperature in heat storage/heat release process, and easy control of heat storage system, and is one of the potential research directions of heat storage technology at present [13–15].

PCMs are functional materials that store and release latent heat through reversible melting and cooling processes. In the past few years, PCMs have been widely used in electronic thermal management, solar thermal storage, industrial waste heat recovery, and off-peak power storage systems [16,17]. According to the phase transition forms, PCMs can be divided into solid-solid, solid-liquid, solid-gas, and liquid-gas PCMs. Solid-liquid phase change materials have shown a broader application prospect in energy storage systems because of their advantages, such as high energy storage density, small volume change rate, and expansive phase change temperature range [18–22]. According to different chemical compositions, solid-liquid PCMs can be divided into inorganic PCMs and organic PCMs [23,24]. Inorganic PCMs show high volume energy storage density, flame retardancy, and thermal conductivity [25]. However, the large-scale application of inorganic PCMs is limited by undercooling and phase separation, which will restrict the timely utilization of instantaneous released energy [26]. Organic PCMs do not have phase separation and undercooling, but their low thermal conductivity leads to a large temperature gradient, so organic PCMs are insensitive to temperature changes [27]. In

addition, flammability, volume expansion, and leakage during phase change are the main problems hindering the development of phase change materials. Therefore, extensive research mainly focuses on the shape-stable PCMs (ss-PCMs) as new composite phase change materials.

SS-PCMs are usually composed of PCMs and porous materials, in which PCMs are used for thermal energy storage, and porous materials are used as shape stabilizers and thermal conductivity enhancers. After adding porous skeleton materials, the molten PCMs are adsorbed in the skeleton due to the capillary action of the skeleton so that the composite PCMs still keep their fixed shapes after energy storage, so they are called stereotyped PCMs [28–30]. The phase change temperature of composite phase change materials can be adjusted by selecting different kinds of phase change materials. With the development of science and technology, porous skeleton materials' types and preparation technology are constantly updated. In the past decades, the porous skeleton materials of ss-PCMs have been extensively studied. According to their research, the porous skeleton materials of ss-PCMs can be divided into four categories: metal foam porous materials, including foam copper-based, foam aluminum-based, and foam nickel-based [31–34]; Organic polymer materials, such as polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyaniline [35–38]; Carbon-based three-dimensional porous materials, including expanded graphite (EG), foamed graphite, foamed carbon, and graphene aerogels [39–42]; Porous ceramic materials, including porous clay mineral ceramic materials (diatomite, expanded perlite, expanded vermiculite), porous oxide and non-oxide ceramic materials (SiO_2 , MgO , SiC) [28,43–46]; As far as we know, there is no comprehensive review of the classification and application of porous material-based composite phase change materials. This paper reviews and discusses the research results of various porous material-based composite phase change materials. The properties of different porous materials are summarized. The effects of different porous materials on thermal conductivity, latent heat, phase transition temperature, undercooling, shape stability, thermal cycle stability, and chemical compatibility of PCMs were comprehensively analyzed. In addition, the applications, advantages, and disadvantages of different porous material-based composite phase change materials in various industries are summarized. Finally, the future research directions of porous composite phase change materials are summarized and prospected.

2. ss-PCMs based on metal foam porous materials

Foam metal refers to unique metal materials containing foam pores. As a metal foam with a continuous skeleton structure, it has a large strength-density ratio, a porosity of over 90%, large pore size (0–15 mm), high specific strength, and relatively high thermal

Table 1
Comparison of thermochemistry, sensible heat and latent heat of phase change.

TES technology	Thermochemical heat storage	Sensible heat storage	latent heat storage
Principle	Based on a reversible chemical reaction, heat energy and chemical energy conversion are realized, and heat storage and heat release are carried out.	Raise or decrease the temperature of the heat storage medium to realize the storage and release of heat ($Q = cm \Delta t$).	The heat storage medium undergoes a phase change process to store and release heat.
Advantages and disadvantages	The energy storage density is the highest, but the design of the heat storage system is complex, the technology maturity is poor, and the one-time investment is enormous.	The heat storage capacity is limited, which is limited by the mass (m) and temperature change (Δt) of the heat storage medium; Simple principle, mature technology, and low cost.	The heat storage density is high, which is related to the phase change enthalpy of the heat storage medium and stores heat at constant temperature; It is in the stage of commercial promotion.
Typical material	Ammoniates, hydrates, metal hydrides, etc.	Water, rock, concrete, liquid metal, etc.	Paraffin wax, salt hydrate, metal compound, etc.

conductivity. Therefore, metal foam-based porous materials have broad application space in improving the system's overall thermal conductivity and relative period [47]. Different types of metal foams have been used as support materials for phase change materials, such as copper foam, aluminum foam, and nickel foam. They have a thermal conductivity of $398 \text{ W m}^{-1} \text{ K}^{-1}$, $237.0 \text{ W m}^{-1} \text{ K}^{-1}$, and $91.4 \text{ W m}^{-1} \text{ K}^{-1}$, respectively.

2.1. Effect of copper foam on the performance of PCMs

Copper foam is a multifunctional material. Because of its porous structure, Copper foam has a large specific surface area. Copper's high thermal conductivity makes it suitable for high heat transfer applications. Copper foam with different porosity and pore density is often used to improve the heat transfer rate of phase change materials.

Wang et al. [48] studied the change of heat storage efficiency of paraffin wax with and without foamed copper and under temperature change. The results show that copper foam can effectively improve the uniformity of heat transfer in paraffin wax and reduce the heat storage time by more than 40%. According to the thermophysical results of paraffin and paraffin/copper foam composite phase change materials, the thermal diffusivity and thermal conductivity of the composite phase change materials increase sharply, showing excellent thermophysical properties. Fig. 1 shows that they used Fluent software to simulate the experiment using solidification/melting and porous zone models. Compared with the experimental results, the finite element analysis software can effectively simulate the phase transformation process of metal foam and paraffin wax and plays a certain role in the design of the phase transformation vessel.

In order to obtain phase change materials with high latent heat and high thermal conductivity, Li et al. prepared copper powder sintered frame/paraffin stabilized phase change materials (CPSF/P-FSPCM) by embedding paraffin in Copper foam by vacuum pouring method [49]. The results show that CPSF/P-FSPCM has good temperature cycle stability, no leakage, and high thermal conductivity. CPSF/P-FSPCM with high porosity contains a large amount of PCM, which can store more sensible heat and latent heat; However, its thermal conductivity is very low. In contrast, low porosity CPSF/P-FSPCM exhibits relatively high thermal conductivity, which indicates that the porosity level of CPSF must be selected according to specific applications.

Similarly, Chang et al. [50] designed and prepared a composite PCM with a stable shape by impregnating organic paraffin wax in graphite-coated copper foam. The preparation process is shown in Fig. 2(A). The modified copper foam keeps super hydrophobic, and the water contact angle is 133.4° . Fig. 2(B) is an SEM image of

graphite-coated copper foam at low and high magnification. It can be seen that the surface of copper foam becomes rough after modification with nano-graphite, which is helpful to adsorb paraffin wax in copper foam. The thermal conductivity of the composites is $2.97 \text{ W m}^{-1} \text{ K}^{-1}$, and there is no paraffin leakage during charging and discharging. Finally, they analyzed the performance of composite materials in solar energy utilization (Fig. 3(C)). Compared with pure paraffin phase change materials, the results show that composite phase change materials can provide higher energy utilization efficiency and longer power generation time for solar thermoelectric systems.

Li et al. prepared a new copper foam/hydrate composite phase change material using modified sodium acetate trihydrate (STA) as inorganic phase change material and copper foam as skeleton support material and compared the effective thermal conductivity of the composite with that of pure STA [51]. The results show that the modified STA has good thermal stability. When the porosity of copper foam is 92.4%, the effective thermal conductivity of SAT composite material containing copper foam is 11 times ($6.8 \text{ W m}^{-1} \text{ K}^{-1}$) of pure SAT composite material. The accumulated heat energy density of the water tank is as high as 467 MJ/m^3 , which is 2.2–2.5 times that of the conventional water tank. In addition, the charging time of composite materials is only 40% of pure SAT. Zhao et al. [52] tested the compatibility of SAT with copper by impregnation corrosion method and prepared composite phase change materials with copper foam as a support matrix. From the SEM images of the inner surface of copper tubes before and after SAT corrosion (Fig. 3), it can be seen that SAT corrosion on copper surface is severe, and a thin corrosion film appears. However, no corrosion film can be observed from the cross-sectional view of the copper sample, which indicates that corrosion has not penetrated the copper material. In terms of performance, low porosity copper foam composite phase change materials have better heat transfer ability under convective conditions. Therefore, although copper foam/SAT composite phase change materials have good thermal stability, low undercooling, and high thermal conductivity, SAT corrosion on copper surfaces is severe.

Jin et al. [53] prepared paraffin/copper foam composites by vacuum infiltration (Fig. 4A). They proposed using the infrared thermal imaging method to realize pore size visualization, further exploring pore size's influence on melting heat transfer of paraffin/copper foam composite phase change materials, and qualitatively and quantitatively studied natural convection, forced convection, and local thermal nonequilibrium effects in pores. The results show that the copper foam samples with 30 PPI and 50 PPI have almost the same melting rate when the wall superheat is 20°C (left in Fig. 4B), and the melting rates of both samples are much faster than those with 15 PPI. This shows that the heat conduction

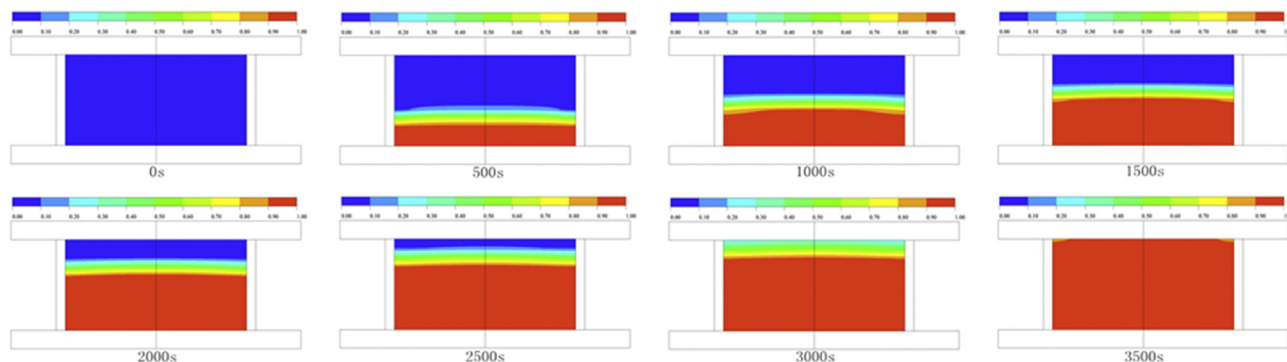


Fig. 1. The melting process of the bubble copper/paraffin at a temperature of 70°C [48]. Copyright 2016, Elsevier.

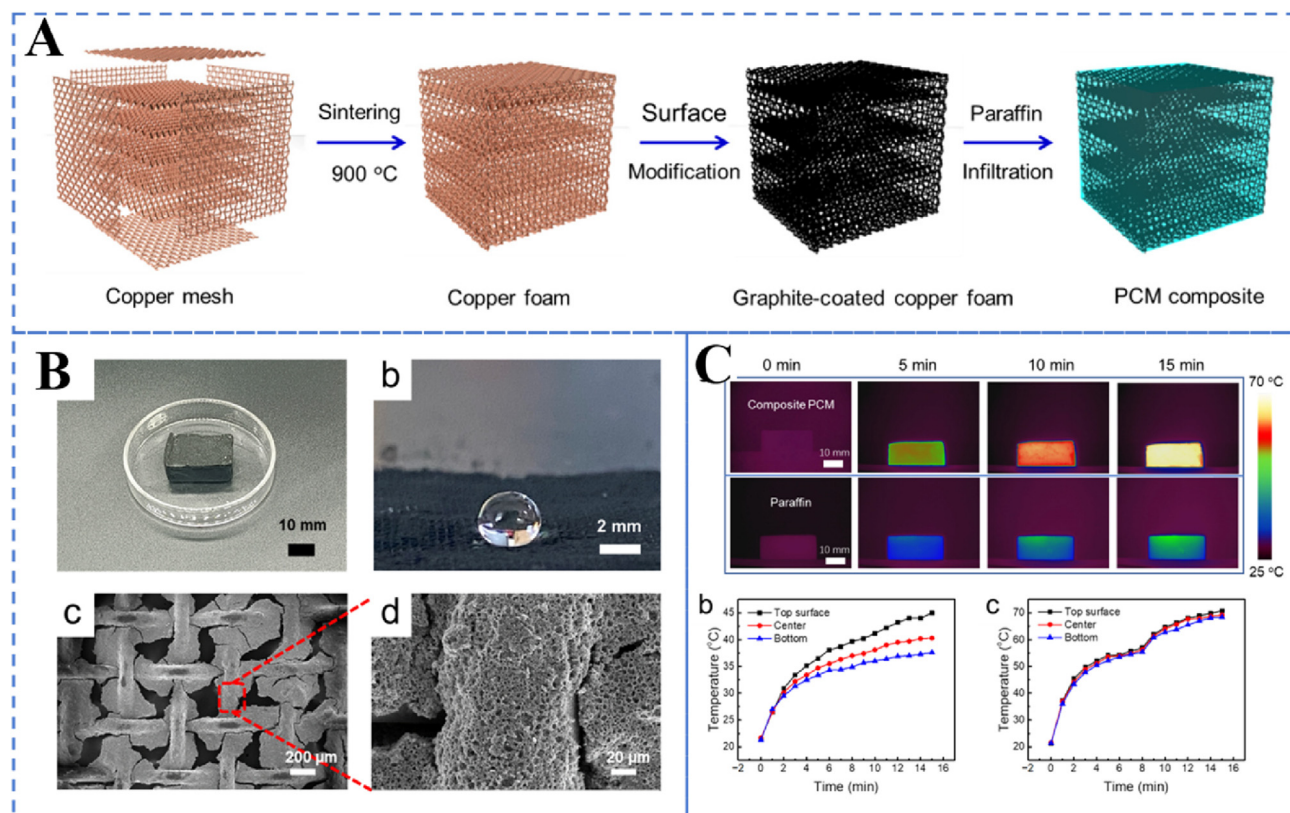


Fig. 2. (A) Fabrication process of the copper foam-based composite PCM. (B) Photographs of the prepared composite PCM and water droplets on graphitized copper foam (above), SEM images of graphite coated copper foam at different magnification (below). (C) Thermal performance of composite PCM during charging [50]. Copyright 2022, MDPI.

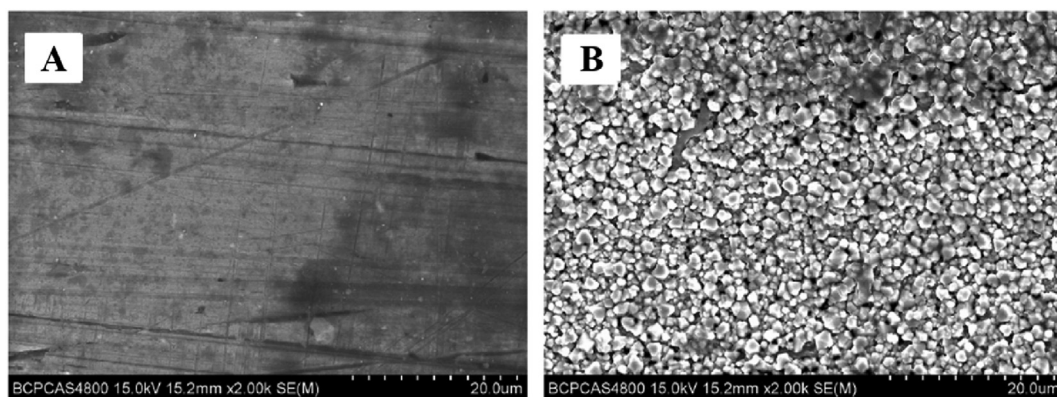


Fig. 3. SEM photos of the inner surface of the copper tube; (A) Before the experiment, (B) After the experiment [52]. Copyright 2018, Elsevier.

enhancement becomes more pronounced with the decrease in pore size. However, when the wall superheats are 30 °C (natural convection is stronger) (right of Fig. 4B), the melting rate of 30 PPI samples is even much faster than that of 50 PPI samples, which indicates that forced natural convection will have negative effects when the pore size becomes smaller. In summary, copper foams with relatively small pores are considered desirable, which may result in faster melting and less pronounced local thermal imbalance effects.

2.2. Effect of aluminum foam on the performance of PCMs

The average thermal conductivity of aluminum foam is 237 W m⁻¹ K⁻¹. Aluminum foam is also widely used to improve the

heat transfer rate of phase change materials. The melting point of aluminum is 660 °C. Therefore, its low melting point limits the application of aluminum foam in high melting salts (carbonates, chlorides, and nitrates).

Zhu et al. prepared low porosity aluminum foam with a porosity of 67%, 75%, and 84% by negative pressure infiltration method and prepared aluminum foam/paraffin composite phase change material with impregnation method paraffin as phase change material [54]. The results show that aluminum foam can inhibit the natural convection during the melting process of pure paraffin wax, and the aluminum foam is heated first, and the heat flow is rapidly transmitted along with the metal ligament structure. Under these conditions, the metal matrix can heat the paraffin wax in the honeycomb unit, and the paraffin wax far away from the heat

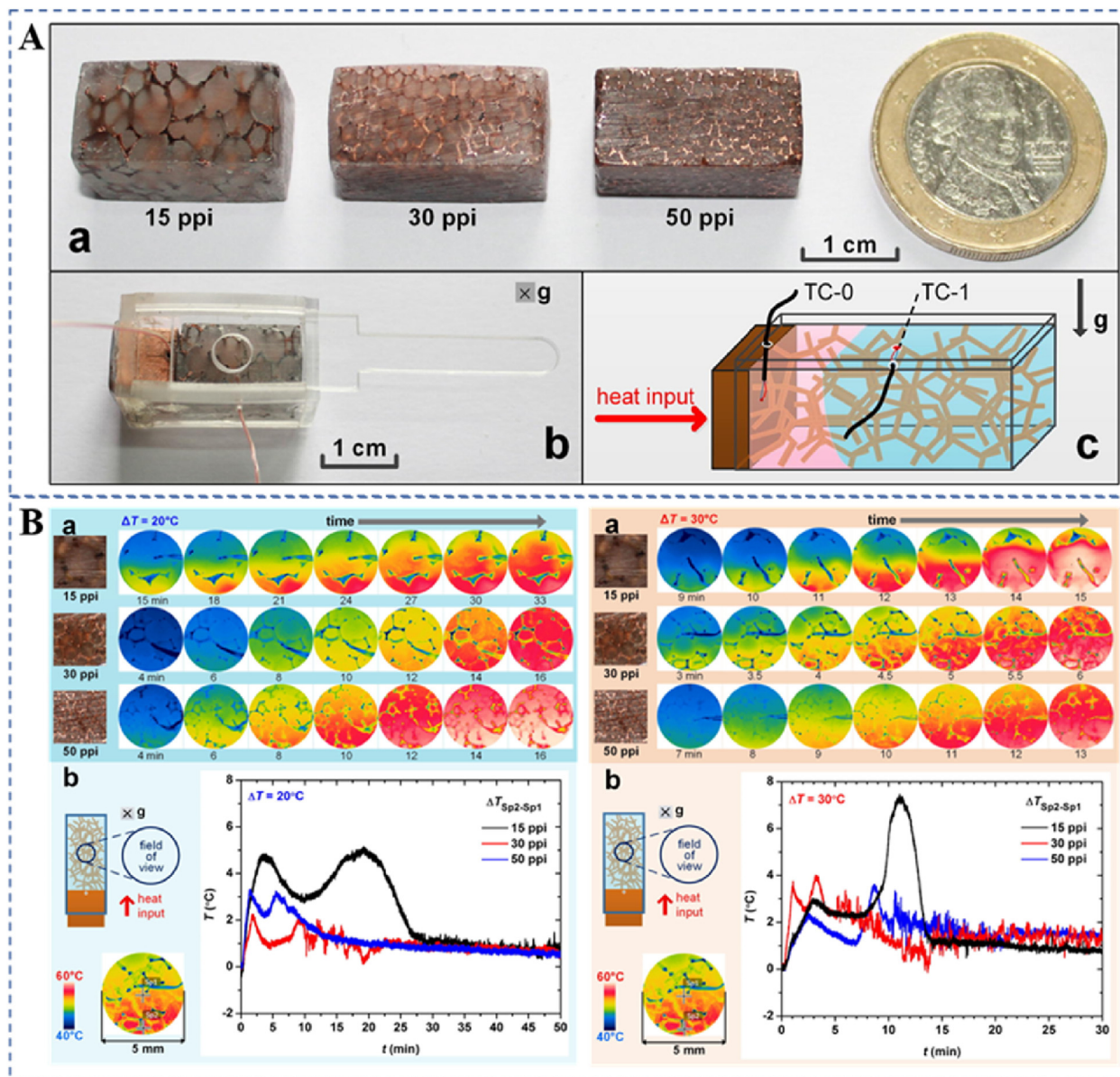


Fig. 4. (A) photographs show: (a) a composite PCM sample prepared by impregnating paraffin wax in foamed copper of three pore sizes, (b) a test section of a melting device, and (c) a schematic diagram of a physical model of the melting of the composite PCM. (B) the evolution of (a) the pore-scale temperature profile and (b) the local temperature difference between two set points in the circular view window during the melting of composite PCM at 20 °C (left) and 30 °C (right) wall superheat was measured by infrared thermal imaging technique [53]. Copyright 2017, Elsevier.

source can be heated rapidly. Therefore, due to the existence of aluminum foam, the thermal conductivity in the composites is enhanced, and the thermal convection effect is weakened. Porosity significantly influences heat conduction and heat convection in the melting process. In composites with a porosity of 84%, the heat flux ratio of convective heat transfer tends to be greater than 20%. With the increase of porosity, the non-equilibrium model has advantages in reflecting natural convection and temperature stratification. With the help of the non-equilibrium model, the characteristics of natural convection in composites can be visualized.

Yousef et al. use paraffin/aluminum foam (AF) composites to thermally adjust photovoltaic (PV) systems [55]. The performance of three PV systems: PV without any modification, PV with paraffin wax as phase change material (PV-PCM), and PV with AF embedded PCM (PV-PCM/AF) were compared and evaluated. The results showed that the electrical efficiency of PV-PCM and PV-PCM/AF systems

increased by 9% and 14%, respectively, in summer compared with unmodified PV. The corresponding enhancement rates were 3.7% and 4.8% in winter, respectively. Through the economic analysis of three kinds of photovoltaics (Table 2), it can be seen that PV-PCM/AF system is the most economical and effective system.

Zhang et al. designed phase change materials with collision energy absorption characteristics, that is, paraffin wax (PW) and expanded graphite (EG) were mixed to obtain EG/PW composites [56]. After that, the EG/PW powder was pressed into the foamed aluminum (Al-f) pores to obtain the EG/PW/Al-f sandwich structure PCM. It is applied to the thermal management and collision performance of batteries. The results show that EG/PW/Al-f PCM has good thermal conductivity, and the thermal conductivity is as high as $4.75 \text{ W m}^{-1} \text{ K}^{-1}$. After holding at 80 °C for 2 h, PCM leaked only 0.8%. When the composite material is applied to the battery for thermal management, the temperature can be well controlled at

Table 2

Makes an economic survey of three photovoltaic configurations [55].

System	P (\$)	CRF (\$)	FAC (\$)	SSF (\$)	S (\$)	ASV (\$)	AMC (\$)	Annual cost (\$)	En,annual (kWh/tear)	Energy Cost Ce (\$/kWh)
PV	30	0.149	4.471	0.069	2.235	0.154	0.447	4.76366	40.99	0.1162
PV-PCM	32.3	0.149	4.813	0.069	2.406	0.166	0.481	5.08124	43.62	0.1165
PV-PCM/AF	33	0.149	4.917	0.069	2.458	0.169	0.491	5.16063	45.07	0.1145

44.4 °C (1 C) and 46.3 °C (1.5 C). Wang et al. [57] studied the heat storage performance of pure paraffin and paraffin/aluminum foam composite phase change materials under different heat fluxes and the influence of composite phase change materials on the cooling performance of lithium batteries. The results show that when the heat flux is 7000 W/m² and 12000 W/m², the phase transition time decreases by 26.4% and 25.6%, respectively. Therefore, the inhibition of natural convection of molten liquid by aluminum foam is more significant than the enhancement of heat conduction. The surface temperature rise of composite phase change materials in lithium batteries is lower than that of pure paraffin wax. Paraffin/aluminum foam composite PCM has an ideal cooling effect in limiting the temperature rise of lithium batteries during discharge.

Zhang et al. [58] prepared paraffin/aluminum foam composite phase change materials by embedding paraffin in aluminum foam with a porosity of 90%. The simulated heat transfer results are compared with the experimental results. The simulation results are in good agreement with the experimental results. In addition, the melting time of aluminum foam/paraffin composite is 26%–28% shorter than that of pure paraffin. Therefore, the aluminum foam skeleton greatly enhances the heat transfer of phase change materials.

2.3. Effect of nickel foam on the performance of PCMs

The thermal conductivity of nickel is relatively low (91.4 W m⁻¹ K⁻¹), but because of its connected pore structure and

high porosity, nickel foam has high aeration, high specific surface area, capillary force, and high melting point (1455 °C). And from an economic point of view, nickel foam is cheaper than copper foam and aluminum foam. Therefore, nickel foam can still be used to prepare composite phase change heat storage materials.

In order to improve the thermal conductivity of hexadecyl palmitate (CP), Wang et al. [59] impregnated it into nickel foams with different pore sizes (70, 90, and 110 PPI) and round (12.7 mm in diameter) by melt infiltration method, and prepared CP/Ni foam CPCMs. TG results show that CP/Ni foam composites have good stability and good compatibility with nickel foam. In addition, the thermal conductivity of CP/Ni foam (70, 90, and 110 PPI) composites increased by 1.88, 2.02, and 4.86 times compared with pure CP, respectively. CP/Ni foam CPCMs exhibits excellent thermal behavior in latent heat storage applications.

Huang et al. Nutmeg alcohol (MA)/nickel foam composite phase change materials were prepared by vacuum melt infiltration [60]. The author observed that MA is well coated with foam nickel (Fig. 5). The effective thermal conductivity of the composites is increased by 1.8 times. The latent heat of phase transition decreased by 3–29%. Because of its relatively low surface area, nickel foam with larger pore size has a lower heat transfer rate. The existence of nickel foam increases the melting point of phase change materials while decreasing the curing temperature.

Hussain et al. also prepared composite phase change materials by coating paraffin wax with graphene layer foam nickel (GcNi)

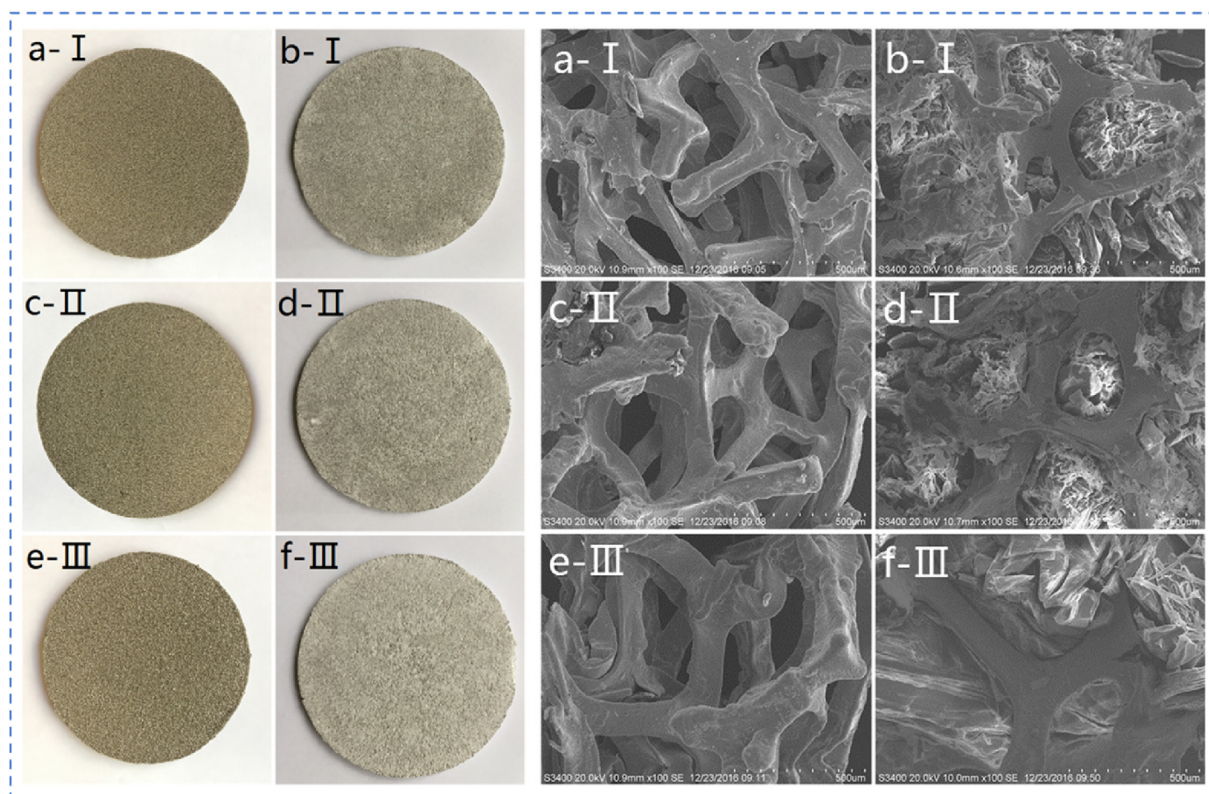


Fig. 5. SEM photographs of nickel foam before and after preparation with different pore size. (I: 40 PPI, II: 70 PPI, III: 90 PPI) [60]. Copyright 2017, Elsevier.

[61]. The results show that the effective thermal conductivity of nickel foam/paraffin and GcNi foam/paraffin composites is 5 times and 23 times higher than that of pure paraffin, respectively. The specific heat and latent heat of GcNi foam/paraffin composites decreased by 34% and 30%, respectively. In addition, the surface temperature of the battery decreases with the decrease of porosity and pore density of nickel foam, but the discharge capacity increases with the increase of porosity and pore density. Liang et al. coated graphene nanosheets onto nickel foam to prepare superoleophilic graphene–nickel foam (GcNi) as a porous support material [62]. And the n-carboxylic acid was adsorbed into the support material by immersion infiltration to form a stable composite. The PCMs can be spontaneously absorbed into the porous support material and remain stable without leakage even above the melting point. The DSC analysis showed that the thermal conductivity of the composite improved 14 times, and the latent heat decreased by 32.7% compared with that of pure paraffin.

2.4. Application of foam metal-based composite PCMs

2.4.1. Thermal management of electronic devices and batteries

In order to increase the life span of electronic devices, the heat flux generated must be rapidly dissipated to the surrounding environment. Tauseef et al. used a copper foam/PCM composite radiator for cooling electronic equipment to observe the relationship between the working time of the radiator and the specific temperature of charging and discharging [63]. The experimental device is shown in Fig. 6. The results show that copper foam/PCM

composites always have better temperature control than pure copper foam. The maximum temperature of RT-35HC/copper foam decreased by 25% at 0.8 kW/m². The maximum cooling rate of RT-44HC/copper foam at 1.6 kW/m² is 24%. The maximum reinforcement rates of RT-35HC/Cu foam and RT-44HC/Cu foam are 8 and 7.7, respectively, at 40 °C and 60 °C.

Baumeister et al. integrate aluminum foam into a PCMs storage system, which can meet the requirements of a high voltage AC system in terms of power, capacity, and temperature range of electric vehicles [64]. The loading and unloading time of the storage system is significantly reduced by introducing foam, and the importance of convection in the overall heat transfer is very low. Compared with the traditional PCM system, the thermal gradient in the foam PCM area is significantly reduced. Therefore, the aluminum foam/PCM storage system meets the requirements of a high voltage AC system. Alipanah et al. [65] made octadecane/aluminum foam composite material into a thermal management system (TMS) for lithium-ion batteries and studied the lithium-ion battery's surface temperature and discharge time by numerical simulation. The results show that TMS made of octadecane/aluminum foam is more effective in storing waste heat generated by batteries. The discharge time was significantly prolonged, and the discharge time was 7.3 times that of pure octadecane. In addition, the surface temperature of TMS batteries using foamed aluminum is more uniform.

Rehman et al. will use foam nickel-PCM-based radiators as electronic cooling systems [66]. The effect of phase change materials in nickel foam was studied, and the performance of nickel

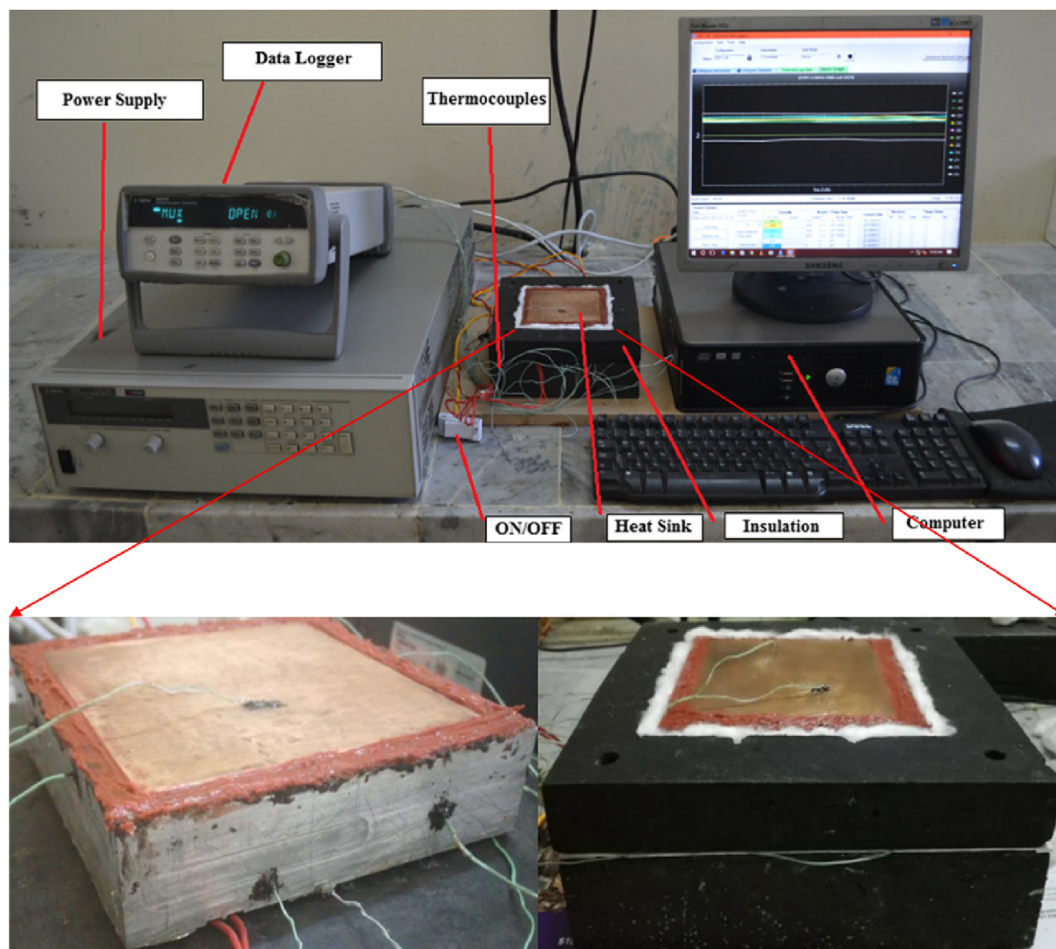


Fig. 6. Experimental device and foam copper/PCM composite radiator [63]. Copyright 2018, Elsevier.

foam flume without phase change materials was compared with that of nickel foam flume without phase change materials. The results show that the base temperature of the radiator decreases significantly after PCM is inserted into foam nickel. And the temperature distribution inside the radiator and at the radiator base is almost the same. Compared with pure PCM, the effective thermal conductivity of foamed nickel-PCM composites increased by 26%. Hussain et al. first proposed and prepared a new type of foam nickel-paraffin composite [67]. The effect of the composite material as a high-performance thermal management source for lithium-ion batteries was studied. The results show that compared with natural air and pure paraffin, the surface temperature of foamed nickel-paraffin composites decreases by about 31% and 24%, respectively. The surface temperature decreases with the decrease of porosity and pore density of nickel foam, but the discharge capacity increases with the increase of porosity and pore density.

2.4.2. Applications in the field of solar energy

Solar energy storage using PCM latent heat is an operationally simple and economically feasible technology that offers renewable energy options in a wide range of applications. Abdulmunem et al. [68] proposed a passive cooling method using the foamed copper matrix (CFM) in photovoltaic panel phase change materials containing multi-walled carbon nanotube additives. The experimental device is shown in Fig. 7. The results show that the addition of CFM reduces the cell temperature by 5.93% and increases the cell efficiency by 5.68% compared with the PV panel using only PCM. Therefore, a multi-walled carbon nanotube foam copper matrix is expected to be used as a passive cooling method to adjust the temperature of photovoltaic cells effectively.

Ejaz et al. used metallic aluminum foam to encapsulate PCM for PV cooling [69]. In this experiment, three different configurations (single PCM, PCM with foam thickness of 8 mm, and PCM with foam thickness of 12 mm) were considered, and the efficiency was checked and compared with the reference module of the naturally cooled photovoltaic panel. Fig. 8 shows the experimental device's temperature distribution, power output, and electrical conversion efficiency and three different types of photovoltaic panels with

RT44 PCM. The results show that the temperature of the PV panel decreases obviously, and the power output and conversion efficiency increase after adding aluminum foam. Compared with matrix phase change materials, the larger the foam thickness, the greater the temperature drop of photovoltaic panels, and the higher the conversion efficiency.

2.5. Summary

The above research shows that metal foam is considered the supporting material to enhance the heat transfer of PCM because of its high thermal conductivity and high specific surface area. PCM/metal foam composite phase change material has a noticeable heat transfer enhancement effect. The most commonly used phase change material for researchers is paraffin wax, which is used to cool electronic equipment and lithium-ion batteries because PCM/metal foam composites reduce the surface temperature. Copper foam is the most widely studied matrix material. Because of its high thermal conductivity, it has a noticeable heat transfer enhancement effect. The solid phase of copper foam composite phase change materials is mainly conductive heat transfer, while the liquid phase is primarily convective heat transfer. Compared with other foam metals, the aluminum foam has the lowest melting point, limiting its use in high-temperature applications. The hot melt of aluminum foam composites decreases with the increase of heat transfer rate and thermal conductivity. Nickel foam composites have excellent performance in cooling applications. The leakage problem of PCM/metal foam composites still needs to be solved. In addition, most studies still regard foam metal as an effective continuous porous medium, and few researchers study the complex thermal behavior at the pore level by numerical or experimental methods. Table 3 summarizes the thermal properties of three PCM/metal foam composites.

3. ss-PCMs based on organic porous polymer materials

Porous polymer is another commonly used carrier for synthesizing shape-stable PCMs. Compared with inorganic porous

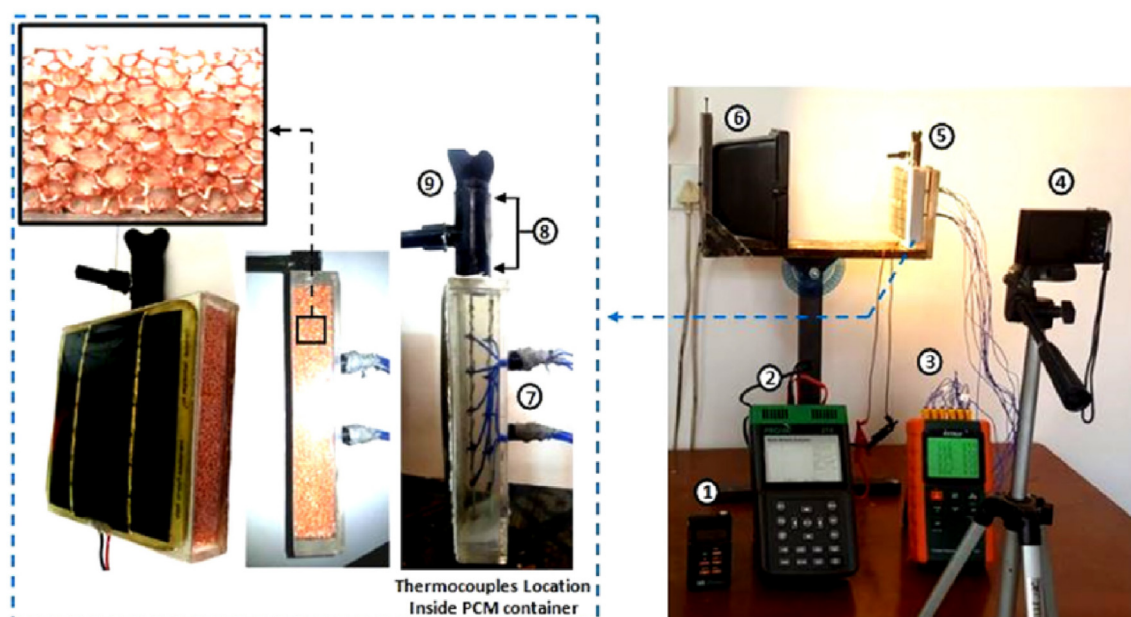


Fig. 7. Actual photographs of the experimental device, 1) Intensity Meter, 2) PV Analyzer, 3) Digital Thermal Recorder, 4) Digital Camera, 5) PV/PCM Model, 6) Solar Simulator, 7) Thermocouple Outlet, 8) Expansion Space, 9) Fill Tube [68]. Copyright 2020, Elsevier.

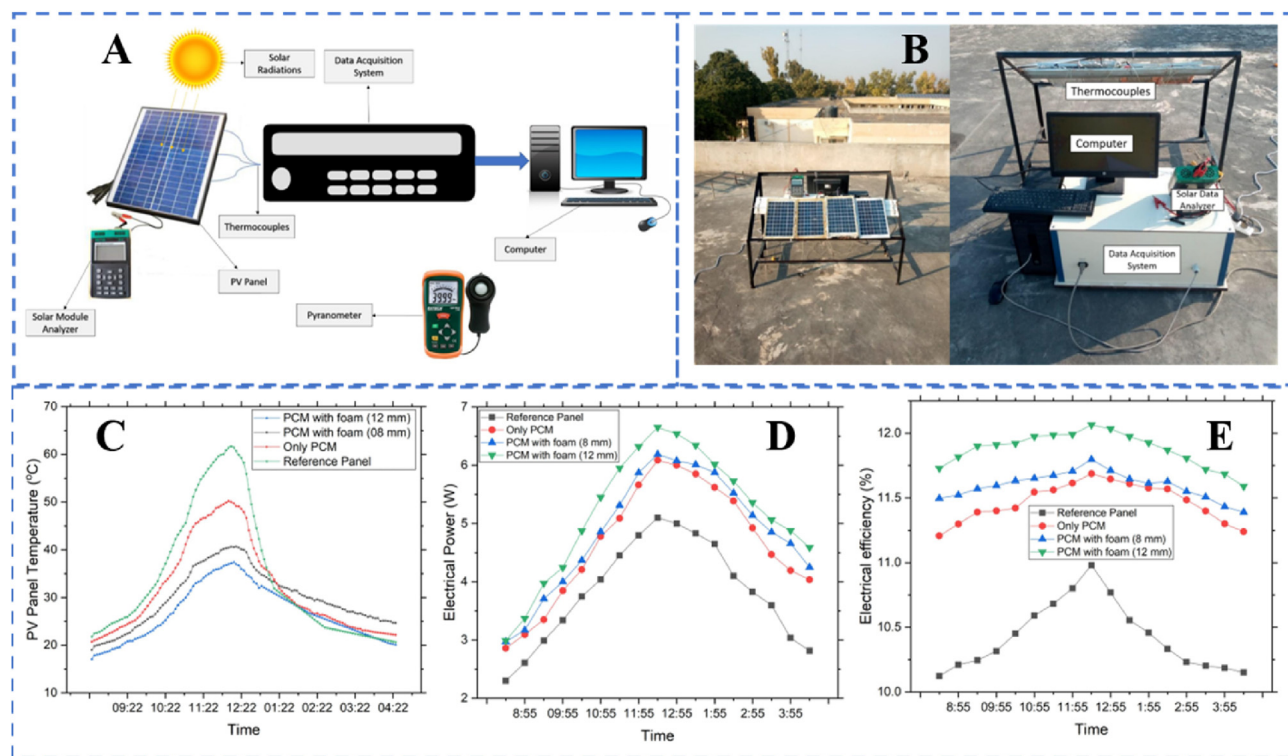


Fig. 8. (A) Schematic diagram of experimental device; (B) Real-time experimental devices; The (C–E) PCM is RT44 for three different types of configurations of PV panels for temperature distribution, power output, and electrical conversion efficiency [69]. Copyright 2022, Elsevier.

materials, porous polymers are composed of light elements such as C, H, O, and N, which have weight advantages. In addition, the surface or skeleton of porous polymer can be easily combined with chemical function, which is difficult for other porous materials [70]. Therefore, porous polymers are expected to adsorb and limit PCMs at low temperatures to prepare ss-PCMs. In recent years, a large number of graded porous polymers with micropores, mesopores, and macropores have been reported, which is expected to improve the application performance of porous polymers [71–76]. In some applications of porous polymers, macropores in polymers serve as storage chambers, mesopores provide transport pathways, and micropores provide adsorption sites for guest species [77–81]. Because of its unique structure, porous polymers may be suitable as support materials to prepare new ss-PCMs with high energy storage capacity.

3.1. Effect of organic porous polymer materials on the performance of PCMs

In order to obtain a new composite phase change material (Fig. 9A), Tang et al. [82] prepared a high pore volume HPP by the condensation reaction of parosaniline base with 1, 4-phthalaldehyde imine. The results show that the graded pores of HPP with high porosity (91.9%) provide enough space for adsorption of PCM (Fig. 9B), the maximum adsorption capacity can reach 75%, and they have good chemical compatibility. The enthalpy of the composite phase change materials is as high as 169.2 J/g, and the enthalpy hardly decreases after 50 thermal cycles. And the undercooling degree of 1-octadecanol is effectively reduced. Similarly, Sari et al. [83] prepared a hierarchical porous foam polymer PHPs using a high internal phase emulsion polymerization method. The synthesized PHPs had graded pores and could absorb up to 75% wt % OD with no leakage behavior. The PHP@CuO/OD composites

were prepared by doping with CuO nanoparticles, and their thermal conductivity was significantly improved. After 500 thermal cycles, the composite PCMs did not exhibit any structural degradation, while the melting-solidification temperature enthalpy remained almost unchanged. They also used the same method to synthesize porous high-internal phase emulsion template polymers (PHPs) with/without carbon nanofiber (CNF) frameworks for stabilizing hexadecanol (HD) as PCMs for thermal energy storage applications [84]. The results showed that HD was retained up to 80 wt% and the composite PCMs had excellent thermal reliability and chemical stability. The thermal conductivity of the composite PCMs was significantly higher than pure HD.

Lu et al. used bio-based polylactic acid (PLA) as supporting matrix material and high-density polyethylene (HDPE) as phase change energy storage material for the first time and prepared a new phase change material with a stable shape by melt blending [37]. The morphology, chemical compatibility, thermal storage performance, shape stability, and thermal stability were studied comprehensively. The results show no chemical reaction between PLA and HDPE in the composite phase change materials, which indicates that they have good chemical compatibility. From the thermal storage performance of PLA/HDPE composites, with the increase of PLA content from 0 wt% to 90.0 wt%, the melting temperature of HDPE decreased from 138.1 °C to 133.5 °C. It is worth noting that PLA can promote the crystallization of HDPE components in blends and improve the energy storage capacity of HDPE components. After 10 thermal cycles, the thermal parameters of the composites remain unchanged.

Babapoor et al. [85] prepared composite phase change materials with stable morphology by single nozzle electrospinning with polyethylene glycol (PEG) as phase change material, polyamide 6 (PA6), and various nanoparticles as supporting materials. The results show that the porous network structure of nanofibers prevents the

Table 3

Summary of work on metal foam/PCM composite phase change materials.

Ref.	Support	PCM	T_m of PCM ($^{\circ}\text{C}$)	C_p of PCM (kJ/Kg-K)	K_{PCM} ($\text{W m}^{-1} \text{K}^{-1}$)	K_{comp} ($\text{W m}^{-1} \text{K}^{-1}$)	ε (%)	H_{PCM} (J/g)	Ω (PPI)	Outcomes
[48]	Copper foam	Paraffin	42.24	1.548–5.012	0.065–0.487	2.879–3.112	97.3	170.4	—	The heat storage time is reduced by more than 40%
[49]	Copper foam	RT-65	57–68	—	—	17.18–156.3	47–74	157.2	—	It can significantly accelerate the heat transfer process and keep the temperature of the equipment within a safe range.
[50]	Copper foam	Paraffin	—	—	—	2.97	92.97	—	—	The thermal conductivity is improved, and there is no leakage during charging and discharging.
[51]	Copper foam	SAT	56.75	1.54	0.59	6.8	92.4	271.1	—	The heat transfer rate increases by 11 times, and the energy density of accumulated heat are 2.2–2.5 times that of a conventional water tank.
[52]	Copper foam	SAT	57.76	—	—	—	88,98	253.6	20	Low porosity copper foam composite phase change materials have good heat transfer ability under convective conditions.
[53]	Copper foam	Paraffin	54.3	—	—	—	95	—	15–50	The smaller pore size of copper foam leads to faster melting and less pronounced local thermal non-equilibrium effects.
[63]	Copper foam	RT-35HC	34–36	2.4	0.2	—	97	200	35	Maximum temperature of electronic devices reduced by 25%
[54]	AF	Paraffin	66.5	2.1	0.2	—	67–84	120.7	—	The thermal conductivity inside the composite is enhanced and the thermal convection effect is weakened.
[55]	AF	Paraffin	43	2	0.2	—	—	165	—	Improved electrical efficiency and reduced power generation costs.
[56]	AF	Paraffin	51.1	—	0.21	4.75	—	233	—	The thermal conductivity of the composites is increased by 22.6 times
[57]	AF	Paraffin	46–52	—	0.21–0.29	—	73–88.9	—	—	The heat storage time is shortened by 26.4%
[58]	AF	Paraffin	52–57	2.8–3.2	0.335–0.558	—	90	200	—	The melting time is shortened by 28%
[65]	AF	Octadecane	—	1.912	—	—	88–97	243.5	—	The discharge time is prolonged, and the surface temperature of the battery is more uniform
[69]	AF	RT44	41–48	2	0.2	—	—	160	—	The temperature of photovoltaic panel decreases obviously, and the power output and power conversion efficiency increase obviously
[59]	NF	CP	52.3	—	0.3432	0.6465, 0.6942, 1.6687	—	180.9	70–110	The thermal conductivity of the composites is increased by 1.88–4.86 times
[60]	NF	MA	36.5–39.5	—	0.1701	0.4768, 0.4010, 0.3229	—	218.4	40–90	The thermal conductivity is increased by 1.8 times. The melting point of phase change materials increases, and the curing temperature decreases.
[62]	NF	Paraffin	38–41	2.3	0.19	4.37	90	143	12.7	The thermal conductivity increases by 23 times; The specific heat and latent heat decrease by 34% and 30%, respectively.
[63]	NF	Paraffin	62.4	2.8	0.162	2.268	96.46	188	—	The thermal coefficient is increased by 14 times; The latent heat decreased by 32.7%.
[66]	NF	RT-44HC	41–44	2	0.21	—	95	256	40	The base temperature of the radiator is decreased by 20.47%, and the effective thermal conductivity is increased by 26%.
[67]	NF	Paraffin	38–41	2	0.2	1	95,97	—	10–30	Compared with natural air and pure paraffin wax, the surface temperature of the battery decreased by about 31% and 24%, respectively.

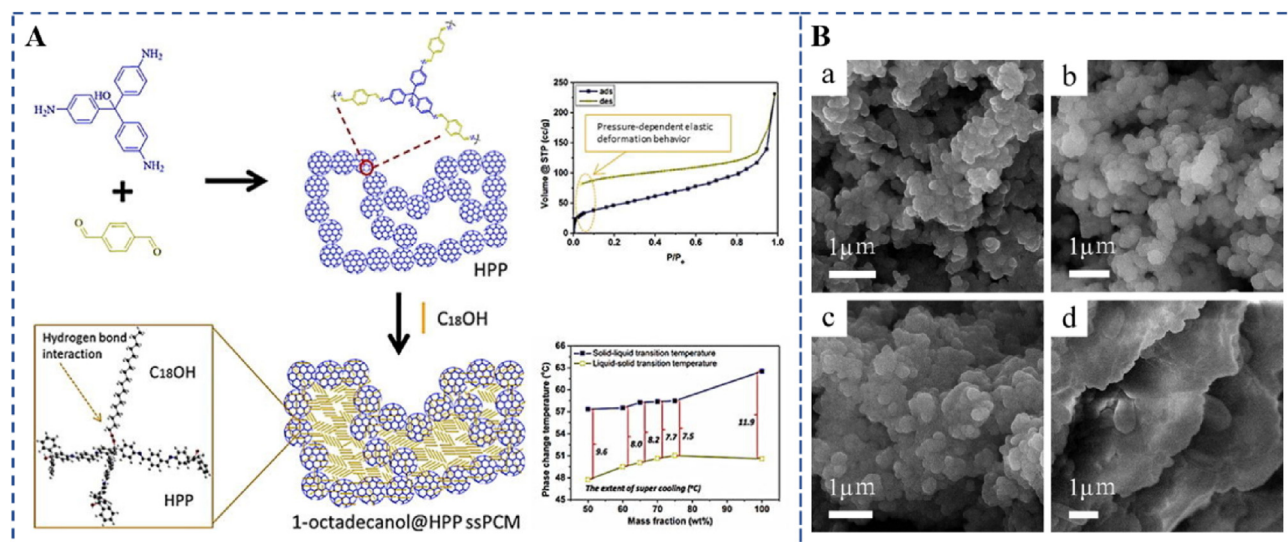


Fig. 9. (A) Synthesis mechanism and thermophysical properties of 1-octadecanol @ HPP ss-PCMs; (B) SEM images of different mass fractions of 1-octadecyl alcohol: (a) 60, (b) 70, (c) 75 and (d) 80 wt% 1-octadecyl alcohol @ HPP ss-PCMs [82]. Copyright 2017, Elsevier.

flow/leakage of molten PEG due to the surface tension and capillary effect between PEG and nanoparticles. According to TGA results, the composite materials have a longer service life at high temperatures. Therefore, the composite fiber has ideal thermal stability. FTIR results show that PEG and PA6 have good chemical compatibility, which indicates that the composites have been formed. Valentini et al. [86] applied salt leaching technology to produce ethylene propadiene monomer (EPDM) rubber foam and TES-EPDM foam containing paraffin. They developed new materials that can store heat energy in building applications. The prepared EPDM foam can adsorb up to 62 wt% paraffin wax. Thermogravimetric analysis shows that the presence of paraffin reduces the thermal resistance of foam, and in any case, the thermal resistance of foam remains much higher than the operating temperature. The foam was heated/cooled 240 times on different substrates. The results showed that the foam did not deform, only a tiny part of paraffin wax was lost, and the residual TES capacity of all samples was higher than 90%.

Wu et al. [87] successfully prepared a series of new side-chain liquid crystal polymers (SCLCP) called P6biCm ($m = 1, 2, 4, 6, 8, 10, 12, 14, 16$, and 18). A new SCLCP/paraffin composite FSPCM was prepared by introducing P6BICM into paraffin wax. The preparation process and physical diagram of the P6biCm/paraffin composite are shown in Fig. 10(A) and (B). Fig. 10(C) is the SEM and POM diagram of composite material. As can be seen from the figure, paraffin wax is evenly wrapped and embedded in the P6biCm three-dimensional network. This network structure can capture paraffin wax and limit the flow of molten paraffin wax. Therefore, even if the ambient temperature is higher than its melting point, the composite FSPCMs remain solid. Fig. 10(D) shows the working reliability of P6BICM/paraffin composites in the long-term phase transformation process by DSC scanning. The results show that P6BICM/paraffin composites have high thermal storage density and excellent thermal stability. The mass percentage of paraffin wax can reach 94 wt%, and the paraffin wax will not leak obviously above the melting point. Because of the good supporting ability of the three-dimensional network of P6biCm, the composite can keep its shape over a long temperature range.

3.2. Application of organic porous polymer-based composite PCMs

3.2.1. Application in the field of construction

Phase change energy storage materials are used in the building field, and the primary purpose is to save energy. Barreneche et al.

[88] developed paraffin/polymer composite phase change energy storage material as a new building material and made an experimental evaluation on strength and sound insulation, pointing out that the developed material can be combined with other materials and applied in building systems.

Chen et al. studied polyethylene/paraffin matrix composites as phase change materials for energy storage in buildings [89]. Paraffin wax is a phase change material, and three types of polyethylene are high-density polyethylene (HDPE), low-density polyethylene (LDPE), and linear low-density polyethylene (LLDPE) are used as structural substrates. Paraffin wax and PE are blended by a parallel and co-rotating twin-screw extruder. Sari et al. [90] think that the polymer-stearic acid blend has great potential in building heating as a stable phase change material for thermal energy storage. In addition, Wang et al. [91] also developed eicosane/polycarbonate composite as a new shape-stable composite phase change material for buildings. Eicosane is added to polycarbonate by solution casting technology. The results show that the blend has good thermal stability and chemical stability.

3.2.2. Application in electricity

Phase change materials can save more than 75% of equipment costs in cooling equipment boxes (rooms) such as communication and electric power. He et al. [92] studied the application of high-temperature phase change materials in latent heat regenerative heat exchangers, which are considered to recover waste heat from industrial furnaces and off-peak power. In addition, Chalc-Sandoval et al. [93] also studied the application of phase change materials encapsulated in polymer matrix utilizing electro-fluid treatment in food preservation. Phase change materials with a temperature of -1.5°C are encapsulated with polycaprolactone, polystyrene, and high impact polystyrene substrates. Polycaprolactone-based composite structure has the best encapsulation efficiency, and 92% of the phase change materials introduced into the polymer substrate are effectively retained.

3.2.3. Applications in the field of solar energy

Shringi et al. [94] have studied phase change materials as energy storage devices for solar dryer drying garlic. The results show that this solar dryer has good energy-saving efficiency in the corresponding drying process. Wang et al. [95] is a new type of composite phase change material with a stable shape used in solar

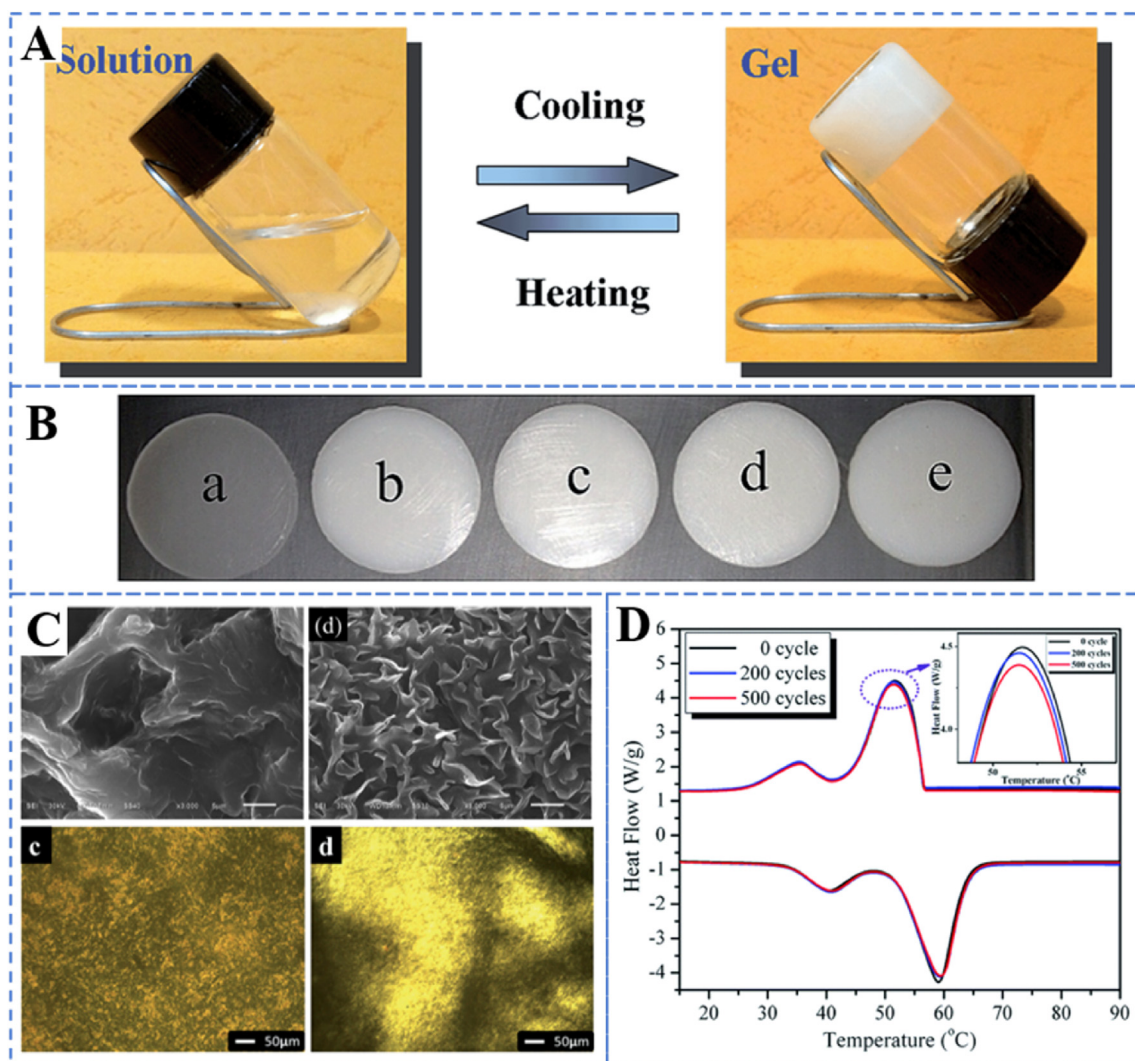


Fig. 10. (A) Preparation process of P6biCm/paraffin composite and (B) physical diagram. (C) SEM and POM of P6biC12/paraffin composites. (D) DSC curves of P6biC12/paraffin composites before and after 200 and 500 thermal cycles [87]. Copyright 2015, Royal Society of Chemistry.

heating and other applications by wrapping SA in the PMMA capsule cavity by UV curing dispersion polymerization. The results show that the composite has good thermal stability and chemical stability. Sari et al. [96] also considered PEG/PMMA blends as new shape-stable phase change materials for heating and ventilation in building solar space. When PEG is used as PCM in the blend, PMMA is used as a supporting material. The prepared materials have a good energy storage effect and have good thermal and chemical stability as phase change materials.

3.3. Summary

The porous polymer matrix is low density, easy to obtain, cheap, non-rusting, and good compatibility with organic phase change materials, and has a high coverage rate. Polymer-based composite phase change materials are easy to process into any shape and have a certain mechanical strength. They are a fast-developing kind of composite-shaped phase change materials in recent years. There are two ways to compound polymer-based and organic phase change materials. One is physical compounding. Physical compounding is to fix the phase change material in the cross-linked network structure of the polymer. There is no chemical bonding between the two; even if the phase change material melts, liquid

phase change material will not occur in macroscopic flow due to the strong van der Waals forces between the molecules. The other is to fix the phase change material on the polymer skeleton with a higher melting point and strength by chemical grafting and blocking, and there is a strong chemical bond between them. However, the low thermal conductivity of the polymer affects the heat transfer and heat exchange rate of the phase change material, as well as the disadvantages of poor high-temperature stability and possible aggregation or decomposition under environmental conditions. Some high thermal conductivity substances such as graphite, carbon fiber, and metal powder can be added to improve the thermal conductivity of polymer-based fixed-form phase change materials. Research shows that graphite is the most effective thermal conductivity enhancer. Therefore, there is still a lot of work to be done to improve the material's thermal conductivity and structural stability to make it widely used. The thermal properties of PCM/porous polymer matrix composites are summarized in Table 4.

4. ss-PCMs based on carbon-based 3D porous materials

Carbon-based three-dimensional porous materials (expanded graphite, foamed graphite, foamed carbon, activated carbon, and

Table 4

Summary of the work of organic polymer/PCM composite phase change materials.

Ref.	Support	PCM	T_m of PCM (°C)	T_f of PCM (°C)	K_{PCM} (W m ⁻¹ K ⁻¹)	K_{comp} (W m ⁻¹ K ⁻¹)	H_{PCM} (J/g)	Loading (wt%)	Outcomes
[82]	HPP	OD	60–65	48–52	—	—	169.1	50–75%	The adsorption capacity of PCM can reach 75%, and the undercooling degree is reduced by 37%
[83]	PHPs	OD	53.65	55.53	0.141	0.153–0.165	261.2	75%	The composites have good chemical compatibility, the adsorption capacity of PCM can reach 75%, and the thermal conductivity is improved.
[84]	PHPs	HD	50.78	52.83	0.141	0.157–0.175	273.2	80%	The composites have good chemical compatibility, the adsorption capacity of PCM can reach 80%, and the thermal conductivity is improved.
[37]	PLA	HDPE	138.1	113.6	—	—	192.2	50%	The composites have good chemical compatibility, and the melting temperature of PCM decreases.
[85]	PA6	PEG	41	—	0.3	0.42	—	33.30%	The composites have good chemical compatibility, and the thermal conductivity is increased by 41.75%
[86]	EPDM	RT-21HC	20.6	12.3	—	—	181.5	62%	After 240 thermal cycles, the composites still show good thermal properties.
[87]	P6biCm	Paraffin	57.2	53	—	—	189.3	94%	Composite materials have high heat storage density and excellent thermal stability.

graphene aerogels) have been successfully used as support matrices to support PCMs. Carbon-based three-dimensional porous materials have highly developed porosity and large specific surface area, which leads to excellent adsorption performance. At the same time, it has good lipophilicity, high-temperature resistance, acid and alkali resistance, and superior heat transfer performance. Compared with foam metal, these materials have the advantages of low price, lightweight, good stability, and compatibility with various PCM. Carbon-based three-dimensional porous materials can be used to adsorb polycrystalline polymers and reduce the shape change during phase transition and improve the mechanical properties, electrical properties, and flame retardant properties of PCMs. Therefore, carbon-based three-dimensional porous materials as supporting materials have broad application prospects in phase change materials.

4.1. Effect of expanded graphite (EG) on the performance of PCMs

EG is a synthetic intercalation compound of graphite, which expands or peels off in a rapidly heated environment. Therefore, it has the characteristics of large pore volume, low density, high thermal conductivity, and high-temperature thermal stability. Therefore, ss-PCM based on EG has good thermal conductivity and relatively high latent heat, which provides a wide range of application fields [97].

Lu et al. [98] heat-treated expanded graphite in a medium temperature box resistance furnace and finally produced expanded graphite with a large number of pores. The ESEM is shown in Fig. 11. It can be seen that expanded graphite with a worm-like structure contains a large number of micropores and mesopores, which can provide a high specific surface area and tremendous capacity to absorb and fix molten paraffin wax. Expanded graphite/paraffin composite phase change materials were prepared by melt blending. The thermal conductivity of the composites was measured. It was found that the thermal conductivity of the composites increased with the increase of the mass fraction of expanded graphite compared with that of pure paraffin wax. The thermal conductivity of paraffin wax/EG80 is more than 4 times that of pure paraffin wax. The preparation process of composite PCM is a pure physical process, and no new substances are produced. Composite phase

change materials can still keep pure paraffin wax's good latent heat storage performance.

Zhou et al. prepared $MgCl_2 \cdot 6H_2O$ /EG based ss-PCM using EG. The hydrophilicity of EG was improved by adding a certain amount of TX-100 [99]. The modified EG was compressed into a round block, and the block was immersed in molten $MgCl_2 \cdot 6H_2O$. The adsorption capacity was as high as 80%, which was 73% higher than that of unmodified EG. The melting point and latent heat of the $MgCl_2 \cdot 6H_2O$ /EG-based ss-PCM composite were similar to $MgCl_2 \cdot 6H_2O$. The latent heat value was as high as 116.7 J/g, and the thermal conductivity was about 7.7 times that of $MgCl_2 \cdot 6H_2O$. In addition, its undercooling degree is about 1.3 °C, which is much lower than that of $MgCl_2 \cdot 6H_2O$ (11.2 °C). After 500 cold and hot cycles, the enthalpy of phase transformation of the composites only decreases by 2.7%. These good properties make $MgCl_2 \cdot 6H_2O$ /EG based ss-PCM have great potential in practical applications. Zou et al. [100] studied the effects of EGs with different particle sizes (50, 80, and 100 mesh) on the thermal properties of $MgCl_2 \cdot 6H_2O$ PCM. The results show that the larger the particle size of EG, the maximum melting temperature, the maximum latent heat of phase transformation, and the minimum undercooling degree of PCM. Therefore, EG-50 should be preferentially used as supporting material to develop shape-stabilized modified $MgCl_2 \cdot 6H_2O$ PCM for cold energy storage.

4.2. Effect of graphite foam and carbon foam on the performance of PCMs

Foam carbon materials are formed by a continuous network graphitized ribbon structure. Their structural properties and thermal conductivity are isotropic, which has better thermal conductivity than traditional carbon fiber reinforced composites and honeycomb aluminum. In addition, the network structure of foam carbon materials increases the contact area between foam carbon materials and phase change materials, which is beneficial to improving the thermal conductivity and mechanical properties of phase change materials. Foam carbon materials can be divided into carbon foam (CF) and graphite foam (GF).

Lan et al. developed an improved vacuum-pressure assisted mixed infiltration technology, which infiltrated mixed alkali salt

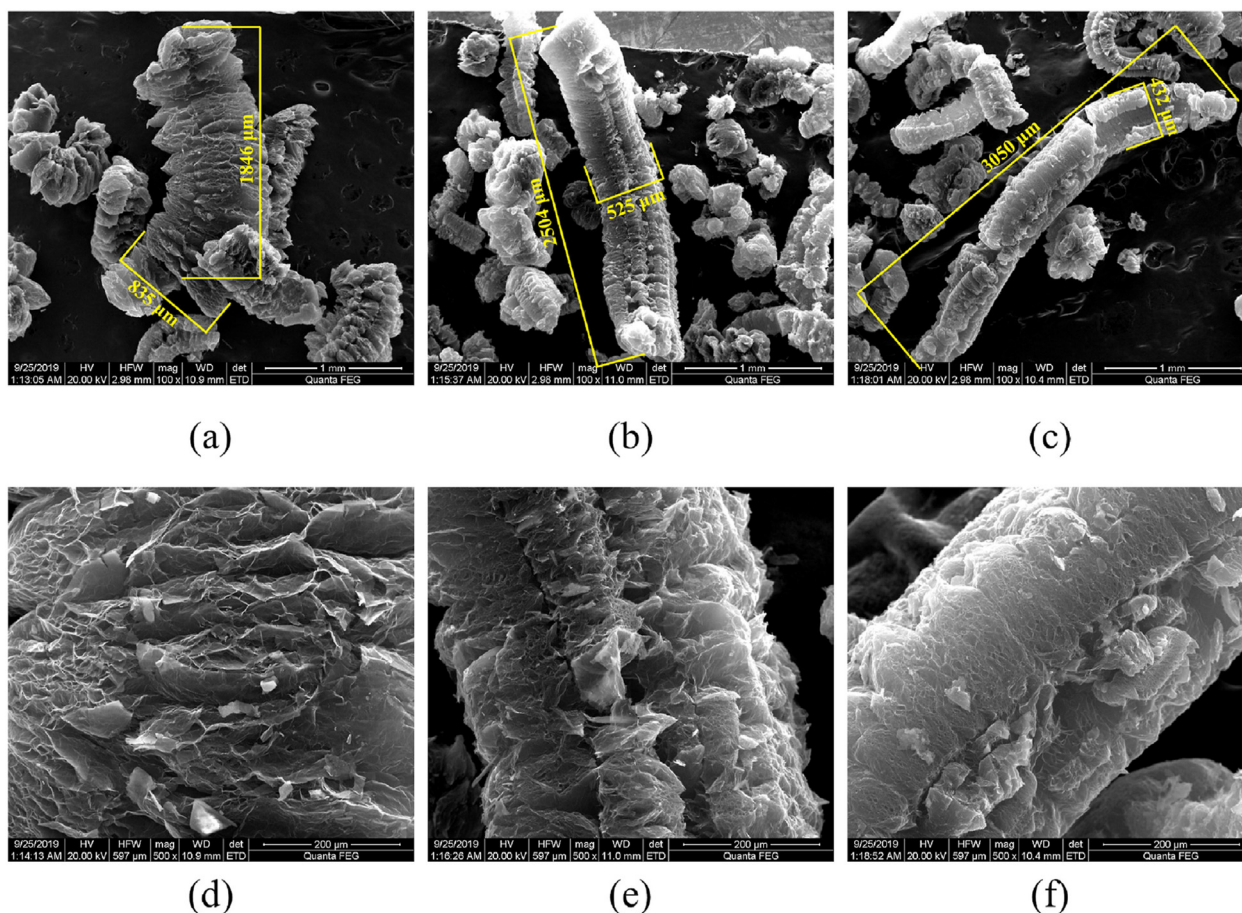


Fig. 11. ESEM micrograph of three kinds of expanded graphite: (a) 50 mesh; (b) 80 items; (c) 100 items. (The magnification is 100); (d) 50 items; (e) 80 items; (f) 100 items. (Magnification 500) [98]. Copyright 2021, Elsevier.

and alkaline chloride salt into high porosity graphite foam as PCM to make composite phase change materials [101]. This technology greatly reduces the installation cost. The results show that the permeability efficiency of PCM is greater than 90% of the available porosity. Compared with pure chloride salt, the thermal conductivity of the composite is increased by more than 40 times. Singh et al. [102] developed a graphite foam/PCM composite in which MgCl_2 is used as PCM, and its melting point is 714°C . The thermal storage properties of two graphite foams with different porosity on the composites were studied. The results show that the liquid phase can be almost filled into graphite foam. And the graphite ligament was not damaged by PCM penetration. At 600°C , the thermal conductivity of the composites is about 30 times higher than that of pure MgCl_2 . TGA-DSC analysis shows that the melting heat of the composites corresponds to the mass ratio of MgCl_2 and graphite foam, and graphite foam has no effect on the melting/freezing temperature of MgCl_2 .

Karthik et al. prepared paraffin/foam graphite composite PCMs by impregnation method. The structural and thermophysical properties of the composites were also investigated [103]. SEM analysis allowed us to conclude that a high percentage of paraffin impregnation (≈ 78 wt%) was achieved without applying any vacuum due to the porous graphite foam's capillary force and surface tension. Storage of the composites at 75°C for 1 month revealed long-term stability and chemical compatibility between paraffin wax and graphite foam. They also tested the thermal properties of the composites. They showed that the thermal conductivity of the paraffin/

foam-graphite composites was 11 and 16 times higher than that of the pure paraffin in the solid and liquid phases, respectively.

Wang et al. [104] synthesized shape-stable composite PCM with paraffin wax as PCM and foam carbon as supporting material. Firstly, melamine foam was heat-treated in a tube furnace, and a new type of carbon foam supporting material was prepared. All the foams maintained a relatively complete and uniform three-dimensional pore structure, and the porosity was over 90%, which was beneficial to the loading of paraffin wax. Only simple physical adsorption occurs between paraffin wax and foam carbon in composites. The thermal properties of the composites were tested. It was found that the thermal storage ability of paraffin wax was not destroyed, and it still showed good thermal properties even after 100 cycles. The research team [105] also prepared lightweight, interconnected porous and high-strength CF by simple pyrolysis method with phenolic resin filled with multi-walled carbon nanotubes as carbon source and polyurethane (PU) foam as a template, which was used to encapsulate polyethylene glycol (PEG) organic PCM. Fig. 12 is the SEM diagram of the prepared sample. It can be seen that the three-dimensional interconnection structure of carbon fiber can ensure that the organic polymer is easily encapsulated in the internal pores, and the adsorption rate is 75%–80%. Almost all latent heat in PEG can be stored or released effectively through a phase transition. The thermal properties of PEG are not affected by the limitation of micropore structure. After 1000 thermal cycles, the composites still show the good shape and thermal stability.

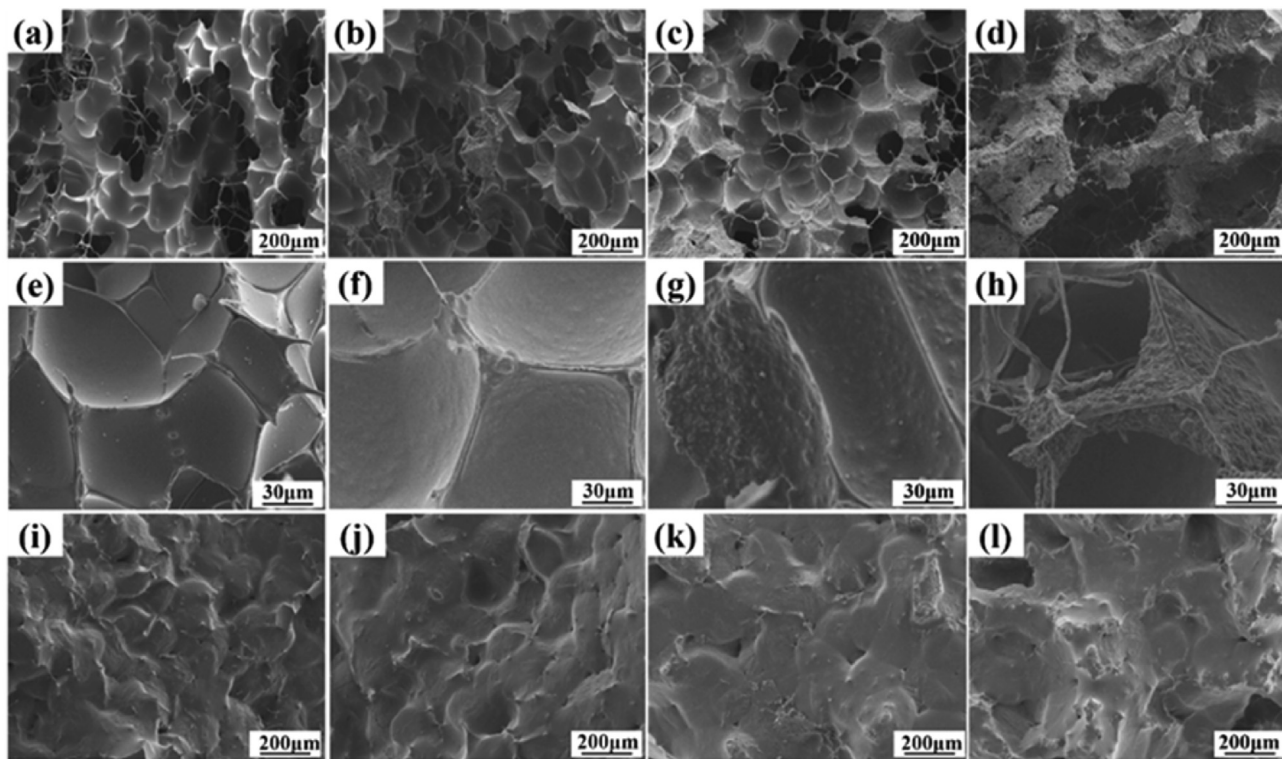


Fig. 12. SEM images of CF, CF2, CF5 and CF10 at (a–d) low magnification and (e–h) high magnification, respectively; (i–l) SEM Images of PEG/CF, PEG/CF2, PEG/CF5 and PEG/CF10 [105]. Copyright 2021, Elsevier.

Biochar materials are promising support matrices for PCMs because of their extensive porous structure, high specific surface area, low density, remarkable adsorption capacity, low cost, environmental friendliness, and relatively high thermal stability and thermal conductivity [106,107]. Biochar is obtained on an industrial scale by pyrolysis from inexpensive biomass or industrial wastes. Therefore, many studies have shown that biochar effectively maintains large amounts of PCMs in vivo without leakage [108]. Hekimoğlu and Sari et al. [106] obtained a carbonized lemon peel (CLP) biochar material by modifying cellulose and pectin substances from the lemon peel with citric acid from lemon juice. And CLP/HD composites were prepared by vacuum using *n*-heptadecane (HD) as PCM. The results showed that the retention of HD in the composite was 65%, and no PCM leakage was detected. The CLP did not have any chemical reaction with the doped HD and had good thermal stability even after 1000 times of cold and heat treatment. The thermal conductivity of the composite PCM was about 77% higher than that of HD. They also solved the genetic defects of different types of organic PCMs and improved their thermal conductivity using low-cost and environmentally friendly AC extracted from spent almond shells (ACAS) [109]. The prepared PCMs/ACAs composites exhibited high thermal degradation properties and cycling stability even after 1200 thaw-freeze cycles. Therefore, the prepared composite PCMs can be used as an economical and environmentally friendly admixture to produce new energy-efficient building materials.

4.3. Effect of graphene aerogel on the performance of PCMs

Graphene is a two-dimensional material composed of carbon atoms with a sp^2 bond. Because of its excellent performance, it shows a lot of fascinating applications. The graphene oxide sheet is self-assembled into a three-dimensional structure by direct growth

of reduction hydrothermal process or CVD method [110], which provides a large number of interconnected open pores and can be filled with liquid PCM to manufacture conductive composite materials with stable shape [111].

In 2013, Zhong et al. used three-dimensional graphene aerogel (GA) as supporting material and octadecanoic acid (OA) as PCM for the first time to prepare OA/GA composite phase change composites [112]. GA can prevent the leakage of molten PCM by capillary force. When the mass fraction of graphene aerogel is 20%, the thermal conductivity of the composites can reach $2.635 \text{ W m}^{-1} \text{ K}^{-1}$, which is 14 times that of pure stearic acid ($0.184 \text{ W m}^{-1} \text{ K}^{-1}$). In addition, due to the high loading capacity of graphene aerogels to octadecanoic acid, the composites show high latent heat (181.8 J/g).

Zhang et al. developed a simple in-situ chemical reduction assembly method to prepare three-dimensional graphene foam and prepared paraffin/graphene foam composites by infiltration [113,114]. Fig. 13A and B are preparation schematic diagrams and microstructure diagrams of paraffin/graphene foam composites. It can be seen that graphene foam presents a three-dimensional network structure with many interconnected and penetrating pores. Due to the interconnected pores of graphene foam and the excellent affinity between hydrocarbon chains of paraffin and graphene sheets, liquid paraffin penetrates into and completely occupies the void space of graphene foam. The addition of graphene sheets enhances the composites' thermal conductivity and solar energy conversion ability.

Graphene aerogels prepared by the conventional hydrothermal method usually comprise a randomly interconnected graphene sheet network. Therefore, the preparation phase change composites are isotropic. To prepare anisotropic phase change composites, Min et al. [115] used anisotropic graphene aerogels (AGAs) by directionally freezing the aqueous suspension of polyamide and GO, then iridizing at 300°C and graphitizing at 2800°C (Fig. 13C).

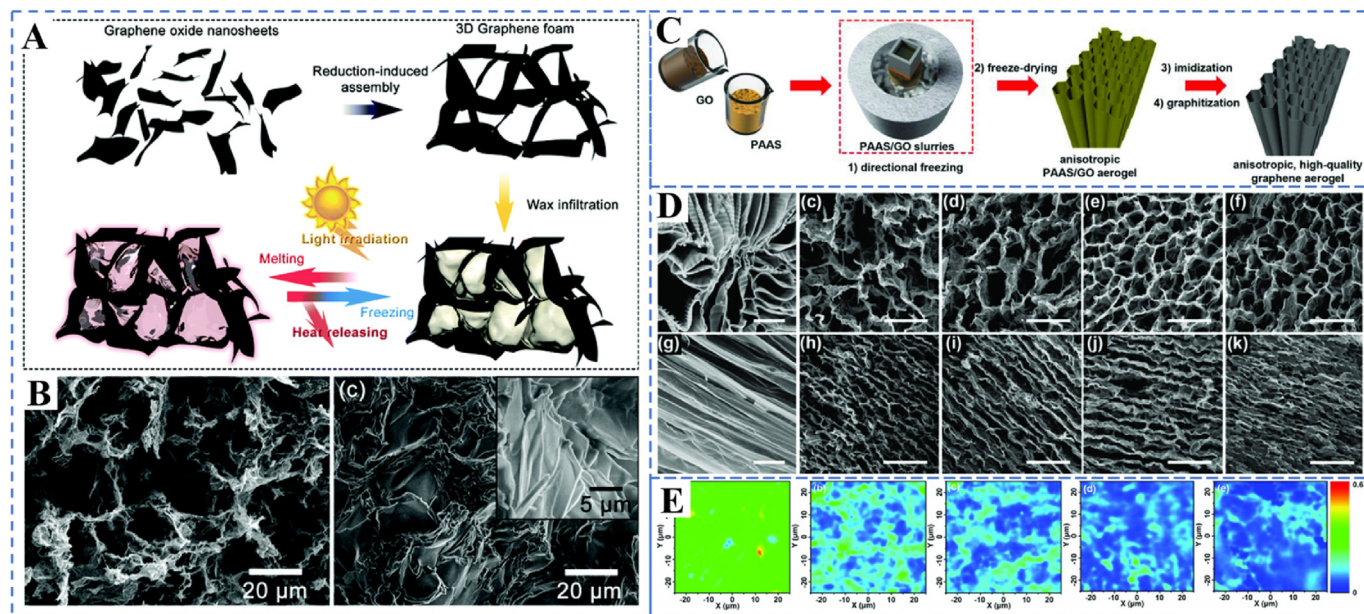


Fig. 13. (A) Schematic diagram of the preparation of paraffin/graphene foam composite. (B) SEM images of graphene foam (left) and paraffin/graphene foam composite (right) [113]. Copyright 2016, Royal Society of Chemistry. (C) Schematic diagram of the preparation of AGAs. (D) SEM images in vertical (top) and horizontal (bottom) views. (E) Raman mapping of AGAs [115]. Copyright 2018, ProQuest.

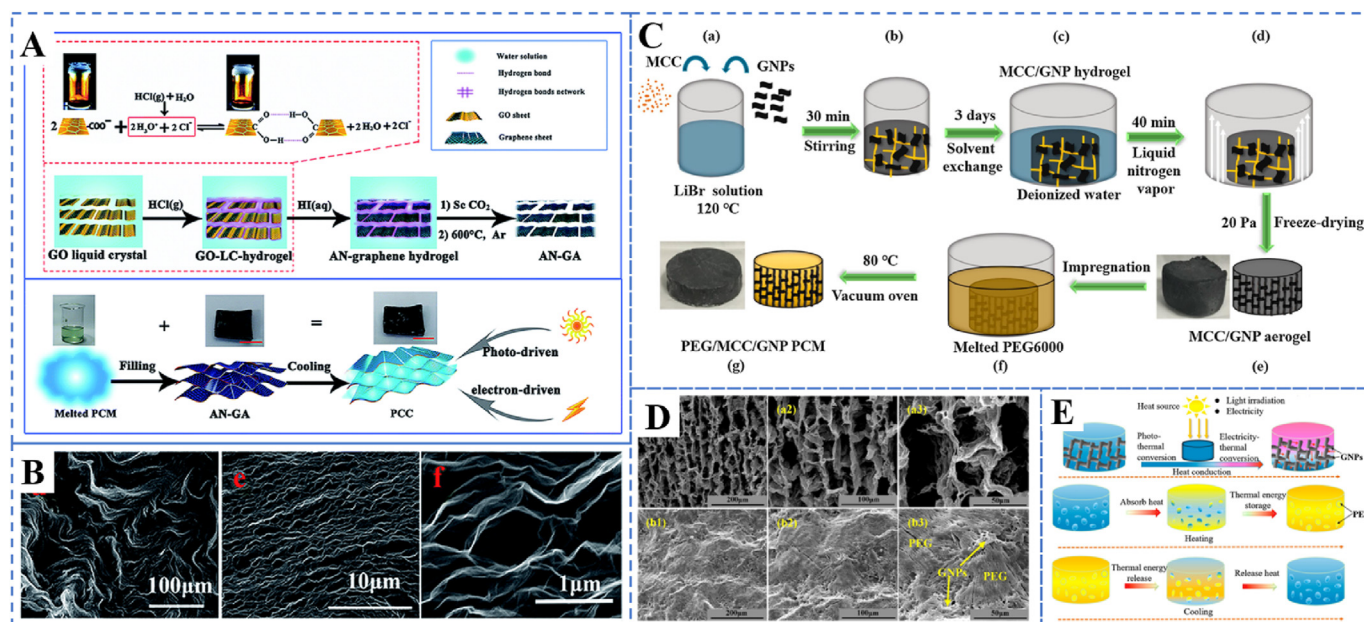


Fig. 14. (A) A schematic diagram of AN-GA and its composite with PCM. (B) SEM images of the AN-GA in the axial direction. (C) Schematic diagram of preparation of MCC/PEG/GNP composite PCMs [116]. Copyright 2017, Royal Society of Chemistry. (D) SEM images of MCC/GNP-1.51 aerogel and MCC/PEG/GNP-1.51 composite PCM. (E) Schematic diagram of energy collection/conversion, storage, and release process mechanism of MCC/PEG/GNP-1.51 composite PCM [117]. Copyright 2019, Elsevier.

After adding GO, apparent anisotropic structure and lamellar to tubular transition were observed in longitudinal and transverse directions (Fig. 13D). In addition, with the increase of GO concentration, the specific surface area and graphitization degree of AGAs also increase (Fig. 13E). Therefore, GO plays an essential role in the design of high-quality AGAs. The transverse thermal conductivity of paraffin/AGAs composite PCMs is $2.68 \text{ W m}^{-1} \text{ K}^{-1}$, and the longitudinal thermal conductivity is $8.87 \text{ W m}^{-1} \text{ K}^{-1}$, which is 24 times higher than that of pure paraffin. This excellent thermal

conductivity is due to the arrangement structure and high specific surface area of AGAs.

Similarly, Li et al. [116] prepared structurally, optically, thermally, and electrically anisotropic graphene aerogels (AN-GA) by curing ordered graphene oxide liquid crystals in situ using gaseous HCl, followed by reduction, supercritical drying, and thermal annealing in inert gases in turn (Fig. 14A and B). The prepared aerogels have high thermal conductivity ($1.71 \pm 0.2 \text{ W m}^{-1} \text{ K}^{-1}$) and electrical conductivity (341.3 S m^{-1}) and have remarkable

anisotropy in thermal conductivity and electrical conductivity. Anisotropic graphene aerogel composites were prepared by infiltrating PCM into the aligned wall structure and confined pores of aerogels. The phase change composite can be driven by applying a small voltage or weak sunlight. Wei et al. [117] prepared microcrystalline cellulose (MCC)/graphene nanosheet (GNP) hydrogel by solution compounding, gelation, and solvent exchange for the first time and successfully obtained porous MCC/GNP aerogel with highly directional stacking of MCC/GNP (Fig. 14C). Fig. 14D shows that aerogels show a three-dimensional orientation framework. The increase of GNPs content is beneficial to the formation of porous structures in MCC/GNP aerogels. When PEG was infiltrated into MCC/GNP, the composite PCMs with 1.51 wt% GNP had higher latent heat (182.6 J/g) and higher thermal conductivity ($1.03 \text{ W m}^{-1} \text{ K}^{-1}$). Compared with MCC/PEG PCM without GNP, the thermal conductivity of MCC/PEG PCM is increased by 23.2%. At the same time, there is almost no latent heat reduction in the heating/cooling cycle, which shows that the composite phase change materials have excellent thermal stability. Finally, they tested the solar energy collection and electric energy conversion of the composite phase change materials, and the principle is shown in Fig. 14E. The results show that the composites have high efficiency in solar energy collection/electric energy conversion, storage, and release.

Besides the hydrothermal method, graphene foam prepared by CVD is also widely used. Qi et al. [118] firstly prepared graphene foam by CVD, then filled the pores of graphene foam with hollow graphene, and finally obtained a new three-dimensional layered graphene foam (HGF). PW/HGF phase change composites were prepared by impregnating molten paraffin (PW) into the porous matrix by vacuum infiltration. The thermal conductivity of PW/HGF composite phase change materials is 74.4% and 87% higher than that of pure paraffin wax and PW/GF composite phase change materials, respectively, and the energy storage density is 95% of that of pure PW. In addition to controlling the microstructure of the scaffold, Tang et al. [119] adjusted the oxidation degree of graphene oxide (GO) before 3D assembly. The results show that with the increase of oxidation level, the graphitization decreases, the types of oxygen-containing functional groups increase, and the higher the oxidation level, the better the shape stability of the composite phase change materials. This difference is probably because the capillary force of porosity and the hydrogen bond between oxygen-containing functional groups of PEG increases with the oxidation degree. Mechanical properties have always been an essential issue in materials science for the practical application of phase change materials. In this respect, Xu et al. [120] characterized the mechanical properties of paraffin and graphene aerogel-based phase change composites by hardness tester and nanoindentation. The results show that the hardness of the composites is 3–7 times that of pure PCM, and the composites can maintain good strength when the melting temperature of phase change materials is higher than that of pure PCM.

4.4. Application of carbon-based three-dimensional porous material-based composite PCMs

Carbon-based three-dimensional porous composite phase change materials have broad application prospects in battery temperature control, building materials, solar energy conversion systems, and industrial waste heat recovery and utilization. From the black appearance of the carbon-based three-dimensional porous conductive scaffold, it can be seen that it has good photo-thermal conversion efficiency and sunlight absorption ability. Therefore, the carbon-based three-dimensional porous conductive scaffold can effectively improve the shape stability and thermal

conductivity of materials and even realize electrical/optical-thermal energy conversion and storage.

4.4.1. Thermal management of electronic equipment and batteries

In order to prolong the life of electronic equipment, the generated heat flow must be quickly dissipated to the surrounding environment. Alrashdan et al. [121] applied bulk paraffin/EG composite PCMs to the thermal management system of lithium batteries and tested the thermal conductivity and mechanical properties of the composite PCM. The results showed that the thermal conductivity tensile strength, compressive strength, and burst strength of EG/paraffin composite PCM increased with PCM content. Gao et al. [122] applied paraffin/EG composite PCMs to the thermal management system of electronic devices. The results show that the heating and cooling efficiency of the chip can be significantly reduced by placing phase change materials in the radiator, and the temperature control time of the radiator can be prolonged; When the heating power of the chip is 20 W, the temperature control time of the radiator filled with composite phase change materials is increased by nearly 20%, thus avoiding the possibility of burning out of electronic devices due to sudden temperature rise and realizing the protection of electronic devices. Huang et al. [123] used the prepared paraffin/EG composite PCMs for thermal management of lithium-ion batteries. The results show that composite PCMs can effectively reduce the heating rate of lithium-ion batteries during operation. When the content of EG is 16–20 wt%, the thermal management efficiency of composite PCMs for lithium-ion batteries is the best.

4.4.2. Applications in the field of solar energy

Zhang et al. studied the application of acetamide as PCM in an active solar energy system [124]. When 10% EG is added to acetamide, the thermal conductivity of composite PCM increases by 5 times. The results show that composite phase change materials' heat storage and heat release rates have been effectively improved. Compared with pure alternating current, latent heat energy storage unit's storage time and regeneration time are shortened by 45% and 78%, respectively. Cao [125] and Zhang et al. [126] prepared low-temperature stearic acid/EG composite PCM and medium-temperature sebacic acid/EG composite PCM. It is found that when the ratio of energy storage materials in composite PCM is 85–90%, composite PCM has higher latent heat. The existence of EG makes the composites have high thermal conductivity and shape stability.

Singh et al. applied graphite foam/PCM to the concentrated solar power station's high-temperature latent heat storage system [127]. They chose magnesium chloride as PCM and simulated three-dimensional heat transfer through commercial software COMSOL. The results show that the addition of foam graphite significantly improves PCM's thermal conductivity and exergy efficiency. After adding graphite foam, the number of pipes in the storage system is reduced considerably than without adding graphite foam. Thereby effectively reducing the production cost.

4.4.3. Application in the field of construction

Li et al. prepared paraffin/EG/diatomite ternary composite PCMs [128]. The porous structure of diatomite and EG can effectively absorb paraffin wax. The mass ratio of paraffin wax to diatomite is 70:22:8. The heating rate of composite PCM was increased by 2.4 times. Then, composite PCM was added to the cement mortar board. The results show that when the content of PCM is 1%, the thermal conductivity of PCM mortar is increased by 32.3%. Zhang et al. [129] Added 10% EG and 90% *n*-octadecane composite PCM into cement mortar to prepare energy storage cement mortar. The results show that the composite PCM has good compatibility with cement mortar. The apparent density of energy storage cement

mortar did not deteriorate obviously. In addition, energy storage cement mortar also plays a role in reducing indoor temperature changes, which is helpful to reduce indoor temperature changes and reduce building energy consumption. Zhang et al. [130] By adjusting the molecular weight of lauric acid and myristic acid and combining them with EG, composite PCMs with high thermal conductivity and good alkali resistance were prepared. Then, PCM-based cement mortar was prepared by adding composite PCMs into cement mortar aggregate. PCM-based mortar has reasonable temperature control and retardation of temperature changes.

4.5. Summary

Carbon-based three-dimensional porous materials have a spatial network structure, which leads to an increase in contact area with PCM. This is beneficial to improving the thermal conductivity, mechanical properties, absorption properties, and leakage resistance of PCMs. It should be noted that carbon materials are easy to oxidize at high temperatures ($>1000^{\circ}\text{C}$), which limits the application range of the composite PCMs. The application research of carbon-based three-dimensional porous composite

Table 5
Summary of work on porous carbon-based/PCM composite phase change materials.

Ref.	Support	PCM	T_m of PCM ($^{\circ}\text{C}$)	T_f of PCM ($^{\circ}\text{C}$)	K_{PCM} ($\text{W m}^{-1} \text{K}^{-1}$)	K_{comp} ($\text{W m}^{-1} \text{K}^{-1}$)	H_{PCM} (J/g)	Loading (wt%)	Outcomes
[98]	EG	Paraffin	61.89	73.49	0.355	1.492	206.24	—	The thermal conductivity is increased by more than 4 times, and PCM's latent heat storage performance can still be maintained.
[99]	EG	MCH	116.69	—	0.465	3.588	144.2	80	The thermal conductivity increases by 7.7 times and PCM's undercooling degree is reduced.
[100]	EG	MCH	11.54	—	0.4544	2.616–3.283	123.1	39.33–56.37	The thermal conductivity is increased by nearly 6 times, and the undercooling degree of PCM is reduced.
[101]	GF	Mixed chloride salt	355	—	1.4–1.8	62.9–101.0	200	90	The thermal conductivity is increased by more than 40 times, and the permeability of PCM is over 90%.
[102]	GF	MgCl_2	710.3	700.4	0.46	43.8	447.8	>97	The thermal conductivity is increased by more than 100 times, and the permeability of PCM is over 90%.
[103]	GF	Paraffin	65	—	0.24	2.6	206	78	The thermal conductivity in the solid and liquid phases increases by 11 times and 16 times, respectively.
[104]	CF	Paraffin	63	54.2	—	—	242.6	97.7	The permeability of PCM is as high as 97%, and it can still show good thermal performance after 100 thermal cycles.
[105]	CF	PEG	67.1	36.4	0.21	0.38–0.68	213.7	75–80	The thermal conductivity increases by 3.24 times, and it still shows good thermal performance after 1000 thermal cycles.
[111]	GA	OD	31.45	18.64	0.1505	1.636	205.4	—	The thermal conductivity is increased, and the thermal stability is still good after 60 thermal cycles.
[112]	GA	OA	56	—	0.184	2.635	186.1	75.00%	The thermal conductivity is increased by 14 times, showing higher latent heat.
[113]	GA	Paraffin	61.8	50.8	0.309	0.617	203	97.80%	The composite material's thermal conductivity and solar energy conversion ability are enhanced.
[115]	AGAs	Paraffin	58	—	0.35	8.87	182.2	98.70%	The longitudinal thermal conductivity of the composite material is increased by 24 times, and it has the potential of solar heat conversion for real-time collection and fast charging.
[116]	GA	Paraffin	54.4	40.75	0.21	1.71	207	94.00%	The composite material has increased thermal conductivity, durability, and high strength.
[117]	GA	PEG	67.6–71.6	27.9–34.4	0.31	1.03	182.9	98.49%	The thermal conductivity of the composite material is improved by 232%, and it has high efficiency in solar energy collection/electric energy conversion, storage, and release.
[118]	HGF	Paraffin	56.5	48	0.27	2.28	160.9	95.20%	The thermal conductivity of the composites is improved by 74.4%, and the photothermal conversion and storage efficiency are high.

phase change materials mainly focuses on heat transfer. Carbon-based composite phase change materials have excellent development potential in conductivity, mechanical properties, and interdisciplinary energy disciplines. The simulation research of carbon-based phase change materials in different application environments needs further development. In addition, some measures should be taken to improve the wettability between carbon materials and molten salt to improve the structural stability and heat transfer performance of the composite PCMs. Related research shows that adding ceramic materials with high surface energy can effectively improve the wettability between carbon materials and high-temperature salts. It is worth noting that there are still few quantitative studies on the relationship between interfacial energy and wettability.

The application research of porous carbon matrix composites PCM has been gradually carried out. Porous carbon-based composite phase change materials have been widely used in buildings, battery temperature control, solar energy, and other fields. However, the application of carbon-based composites in buildings is still less. The reason is high cost and low mechanical properties. Further efforts should be made to study the further improvement of its performance and its application in architecture. Table 5 summarizes the thermal properties of porous carbon matrix composites.

5. ss-PCMs based on porous ceramic materials

Compared with foam metal, porous ceramic materials generally have the characteristics of low density, low preparation cost, high strength, and corrosion resistance, which makes it possible for PCMs to be used in civil fields such as building insulation materials. In addition, compared with other types of composite PCMs, ceramic-based composite PCMs have the following advantages.

- (1) Low skeleton density and high energy storage density of composite materials;
- (2) It can be directly contacted with the heating body, and the heat exchange efficiency is high;
- (3) It can effectively reduce the design cost of the system;
- (4) It can be directly processed and formed according to the actual situation, which is convenient for industrial installation;
- (5) The requirements for packaging containers are low, or no packaging containers are needed, and a large amount of molten phase change materials will not leak.

Therefore, ceramic matrix composite phase change materials have become the research hotspot of energy storage materials in recent years.

5.1. Effect of porous clay mineral ceramic materials on the performance of PCMs

Most clay mineral materials (diatomite, expanded vermiculite, expanded perlite, etc.) have a porous structure, large specific surface area, high thermal stability, and good chemical compatibility [131,132]. PCM and clay mineral materials can be effectively loaded by capillary force, surface tension, hydrogen bond, van der Waals force, etc. The leakage can be prevented by vacuum impregnation and melt adsorption. Therefore, clay minerals are considered an ideal skeleton material for composite phase change materials.

5.1.1. Diatomite

The mineral composition of diatomite is mainly opal and its varieties, and its structure contains large micron/nanometer pores [133], and the porosity can be as high as 80%–90% [134]. Its unique porous structure is shown in Fig. 15 [135]. The above characteristics determine that diatomite can better adsorb and seal PCM and endow PCM with good mechanical properties. At the same time, it has the characteristics of abundant reserves, easy access, and low price. At the same time, diatomite also has good corrosion resistance and high-temperature resistance [136]. Therefore, it has been widely studied as an excellent carrier for PCM packaging with low cost and high performance.

Liu et al. [137] studied the energy storage properties of phase change composites prepared by impregnating paraffin wax with diatomite by mixed grinding and heating. The research results show that. Paraffin wax is effectively impregnated into the pores of diatomite, and the rest of the paraffin wax covers the surface of diatomite. Through FT-IR spectrum analysis, it is found that they have good chemical compatibility. TGA-DSC results show that phase change materials have high thermal stability. Because of the existence of diatomite, the thermal conductivity of gypsum-based phase change composites decreases compared with pure paraffin wax, while the phase change temperature increases. This shows that the phase change composites have the original paraffin wax's high phase latent heat characteristics. After three thermal cycles, there is no effect on the chemical structure of the composites, which indicates that the composites are stable.

Diatomite can be used to support high-temperature PCMs because of its good corrosion resistance and high-temperature resistance. Deng et al. [134] selected KNO_3 as high-temperature PCM and diatomite as the carrier to prepare KNO_3 /diatomite composite phase change material using a mixed sintering method (Fig. 16). They evaluated its thermal properties, physical properties, and chemical compatibility in detail. The results show that when the diatomite ratio is 40–55 wt%, the appearance of the composite is perfect, and there is no sign of salt leakage. The optimum

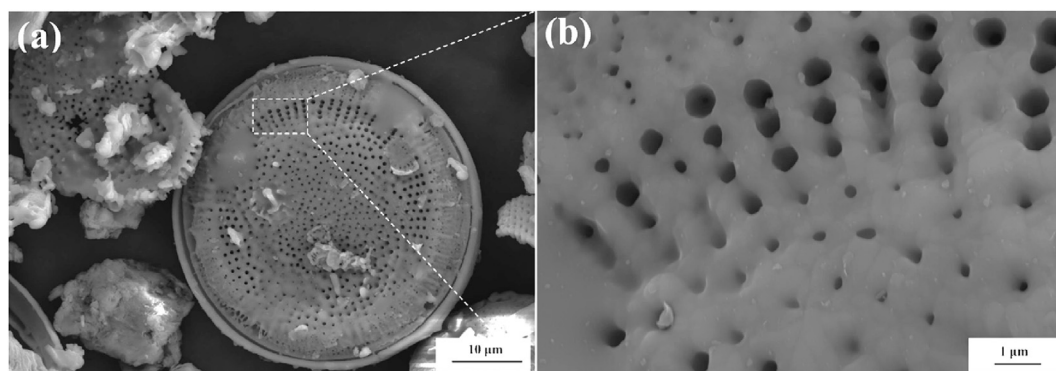


Fig. 15. Microstructure diagram of diatomite [135]. Copyright 2022, Elsevier.

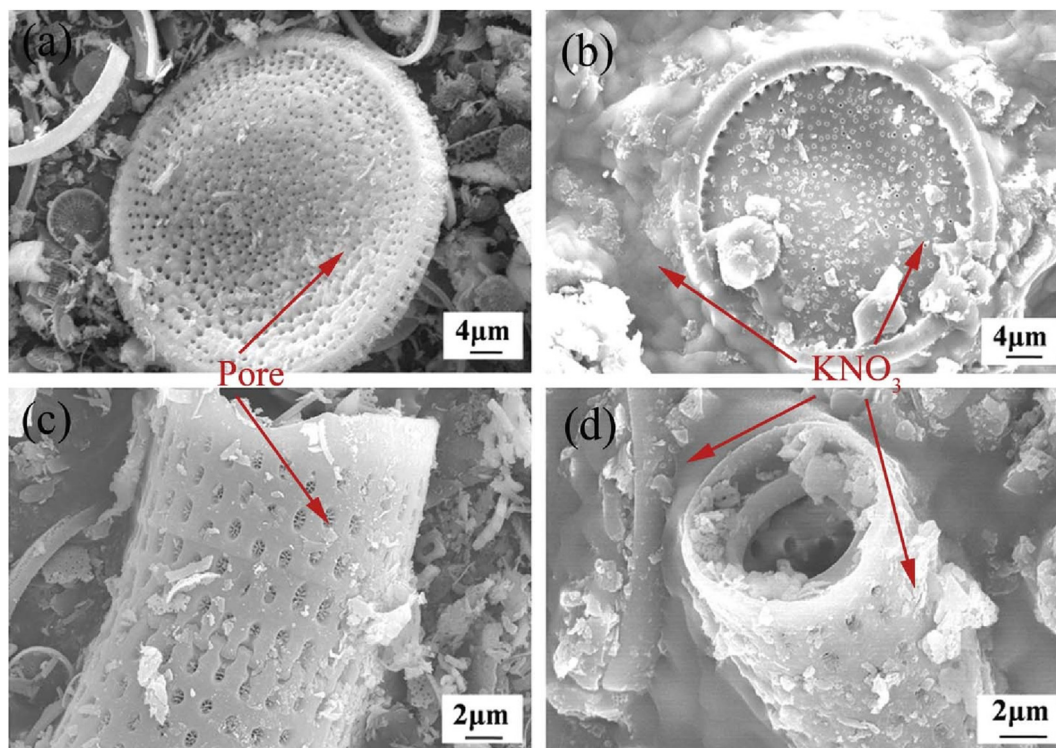


Fig. 16. SEM images of diatomaceous earth (a, c) and ss-CPCM containing 65% KNO_3 (b, d) [134]. Copyright 2017, Elsevier.

absorptivity of the composites is 65 wt%. It has good chemical compatibility with diatomite. After 50 thermal cycles, the weight loss rate of composite phase change materials is only 4.1%, and the melting and freezing phase change temperatures have little change. Therefore, KNO_3 /diatomite ss-CPCM can be a potential storage medium for thermal storage in solar power plants.

Wen et al. [138] A new type of decanoic acid-lauric acid (CA-LA) eutectic mixture as phase change material (PCM), diatomite as support material, was prepared by vacuum impregnation method with stable morphology of CA-LA/diatomite composites PCM. The results show no chemical reaction between CA-LA and diatomite, only physical interaction. Due to the porous structure of diatomite, CA-LA is adsorbed in large quantities and can be prevented from leaking. The maximum adsorption capacity of CA-LA in diatomite is 53.6%. After 200 heating and freezing cycles (Fig. 20), CA-LA/diatomite composites have good chemical structure and thermal stability. Therefore, based on the chemical and thermal stability of CA-LA/diatomite composites, CA-LA/diatomite composites have great potential in building applications.

5.1.2. Expanded vermiculite (EV)

As a natural compound in concrete, Vermiculite has evident expansion after high-temperature treatment. Expanded Vermiculite (EV) has a unique porous structure, low density, fire resistance, and non-toxic properties. Therefore, expanded Vermiculite is also suitable as the skeleton material of shaped phase change materials.

Zhang et al. [139] doped capric acid-palmitic acid-stearic acid (CA-PA-SA) eutectic mixture into EV by the vacuum impregnation method and prepared phase change composites with a stable shape. CA-PA-SA is used as thermal energy storage material, and EV is used as supporting material. SEM results show (Fig. 17) that the porous network of EV fully absorbs CA-PA-SA. Because it is absorbed by capillary and surface tension, the leakage of molten CA-PA-SA can be prevented even when heated to about 70 °C. 70% of CA-

PA-SA/EV composite phase change materials melted at 19.3 °C, and the latent heat value was 117.6 J/g. After 200 cold and hot cycles, the composite phase change materials still had a high latent heat value. Besides, after adding 5% copper powder, the thermal conductivity of CA-PA-SA/EV could increase by about 49.58%.

Similarly, EV can also be used as a support for high-temperature PCMs. Li et al. [140] NaNO_3 /EV composite phase change materials with stable morphology were prepared by the direct impregnation method. The results show that the adsorption capacity of the EV pair is as high as 734.6%, and the EV skeleton is only physically combined with NaNO_3 salt without any chemical interaction. The thermal conductivity of composites is higher than that of pure sodium nitrate. Wen et al. [141] Lauric acid (LA)/expanded vermiculite stabilized phase change materials were synthesized using the vacuum impregnation method. XRD and FT-IR results show that lauric acid and expanded vermiculite in the composites do not react chemically but only physically combine. After 200 thermal cycles, the enthalpy of phase transformation of the composites has no apparent change.

5.1.3. Expanded perlite (EP)

Perlite is an acidic vitreous rock originating from volcanic eruptions [142]. When the temperature of perlite is heated above 870 °C, its volume will expand to about 4–20 times the original volume. After cooling, expanded perlite (EP) with a honeycomb structure will be obtained. EP has a loose and porous internal structure (Fig. 18) [131,143], so it has a strong adsorption capacity by virtue of the capillary force and surface tension provided by its porous structure.

Li et al. [131] NaNO_3 /EP composite phase change materials were prepared by mechanical stirring and direct melt adsorption. In the composites, EP as skeleton material has good thermal stability in the range of 300–900 °C and only physically binds with NaNO_3 as PCM. SEM (Fig. 19) clearly shows the capillary force and surface

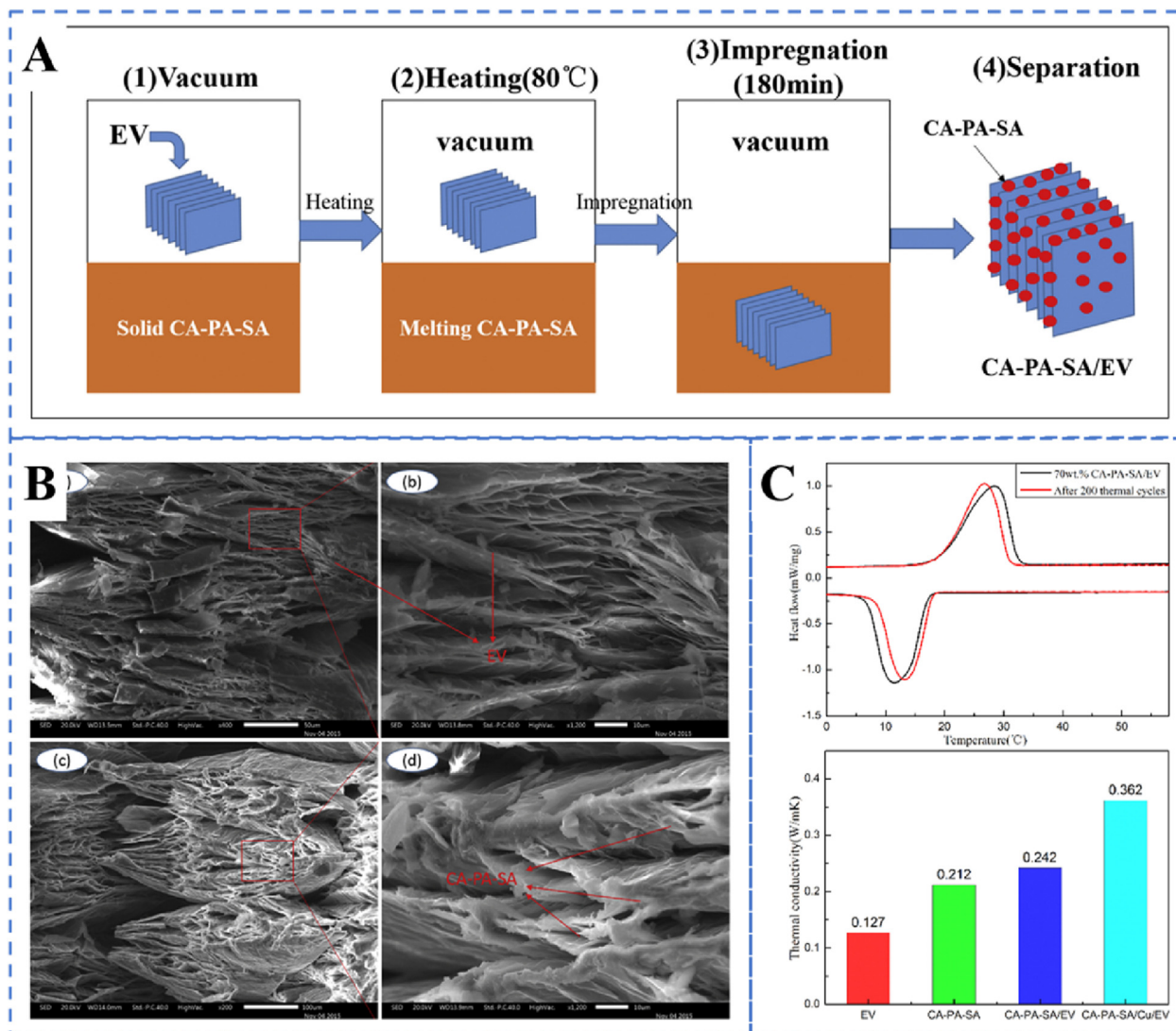


Fig. 17. (A) Schematic diagram of CA-PA-SA/EV composite prepared by vacuum impregnation method. (B) SEM images of EV (top) and composite PCM (70 wt% CA-PA-SA/EV) (bottom). (C) DSC curves of composite PCM and thermal conductivity of various materials before and after 200 thermal cycles [139]. Copyright 2017, Elsevier.

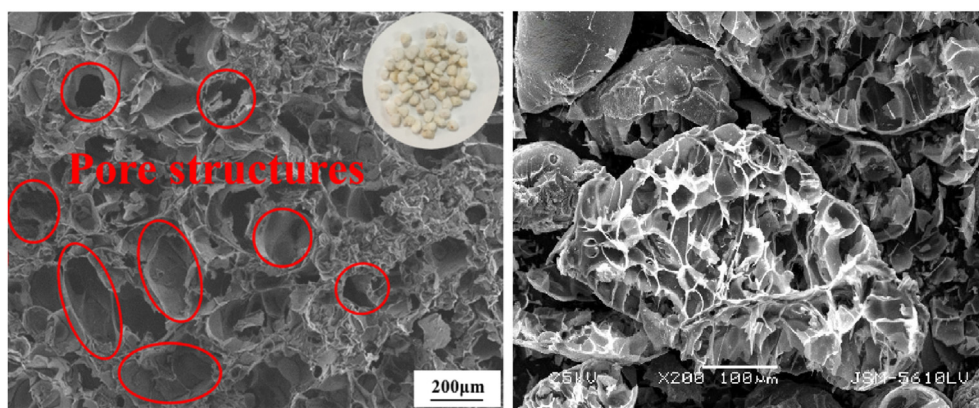


Fig. 18. SEM micrograph and photograph of EP [131,143]. Copyright 2016, Elsevier.

tension provided by the porous structure of EP, which can effectively adsorb molten NaNO_3 without leakage. The thermophysical properties of the composites show that the composites have good

thermal conductivity. Compared with the thermal conductivity of pure sodium nitrate ($0.5 \text{ W m}^{-1} \text{ K}^{-1}$), the thermal conductivity of pure sodium nitrate is $0.84 \text{ W m}^{-1} \text{ K}^{-1}$ at 25°C when the content of

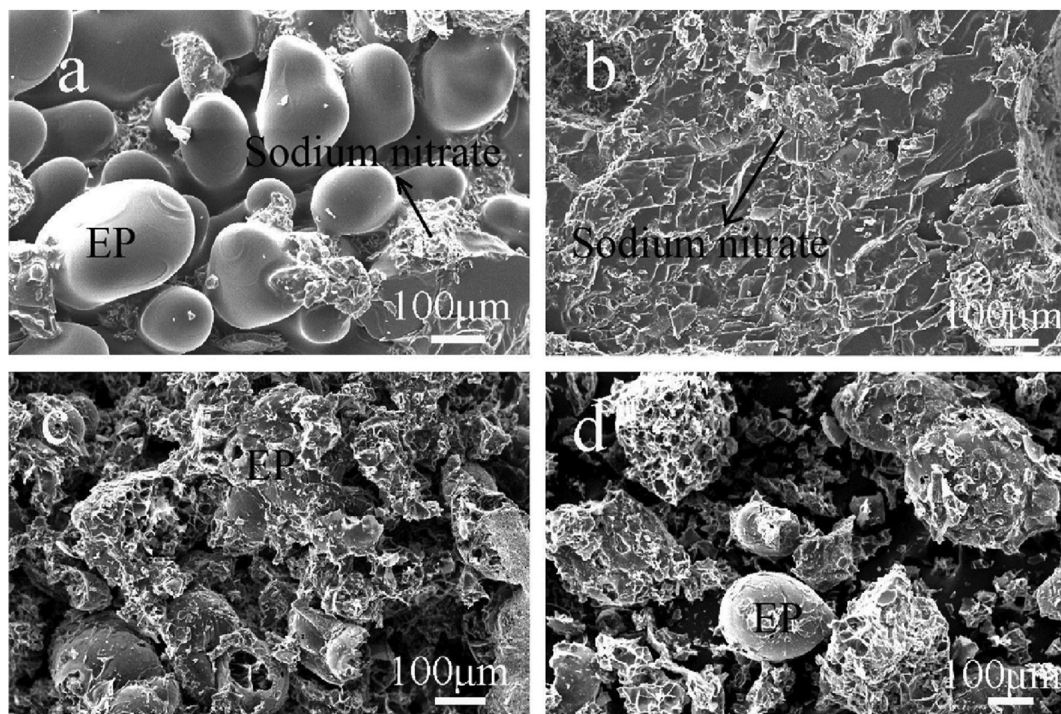


Fig. 19. SEM diagram of composites: EP content (a) 10%; (b) 20%; (c) 30%; (d) 40% [131]. Copyright 2016, Elsevier.

EP is 10%. It shows that sodium nitrate and EP composites can improve the thermal conductivity of the materials.

Hasanabadi et al. [144] impregnated paraffin wax (PA) as phase change material into expanded perlite (EP) particles by vacuum impregnation method. To prevent PCM leakage, PA/EP composites were coated with polystyrene. It is found that EP particles adsorb paraffin wax in their porosity by vacuum impregnation, which is twice their mass. After 300 melting-freezing cycles, the phase transition temperature and latent heat of PA/EP composites decreased by 0.96% and 2.3%, respectively. In addition, no paraffin leakage was observed after the thermal cycling test. Mekaddem et al. [145] also prepared PA/EP composites by the impregnation method and mixed the composites with gypsum to obtain heat-adjusted gypsum composites. FT-IR and DSC characterized the thermal properties of the composites. The results show that the composite components have good chemical compatibility, thermal reliability, and stability. No change in phase transition and transition temperature was observed.

5.1.4. Other clay mineral materials

Other clay minerals include silica fume, pumice, kaolin and bentonite [146–153]. All of them can generate porous skeletons, which can provide loading space for PCMs. The porous structure of composite phase change materials with these materials as skeletons can effectively prevent the leakage of inorganic salts by virtue of their capillary forces and surface tension. In addition, such materials usually have good high-temperature resistance and excellent chemical compatibility with inorganic salts. Therefore, the above materials are also well suited for use as skeleton materials for shaped phase change materials.

PCMs can effectively load silica fume (RSF) into the pore structure of RSF due to its volcanic ash activity and absorption capacity. Sari et al. [146] prepared high thermal performance and low-cost composites by direct impregnation method using RSF as the carrier material and PEG as PCM. They used CNF-CNT-GNP as an ultra-high

thermal conductivity additive, which can significantly improve the thermal conductivity of RSF/PEG composites in terms of thermal conductivity. They also prepared CA-PA/RSF composites by vacuum impregnation method using CA-PA as PCMs, which can be considered a more promising passive solar thermal conditioning composite for building envelopes [147]. Then they prepared a series of composite phase change materials using kaolin, bentonite and pumice as support materials [148–153]. The results showed that the composite phase change materials have good thermal outperformance.

5.2. Effect of porous oxide and non-oxide ceramic materials on the performance of PCMs

In addition to the common porous mineral ceramic-based materials mentioned above, some porous oxide and non-oxide materials (SiO_2 , MgO , Al_2O_3 , SiC) have good chemical compatibility, a large specific surface area, and good stability. They can be used as excellent carriers for PCMs.

5.2.1. Effect of SiO_2 -based porous materials on the performance of PCMs

SiO_2 is an important inorganic amorphous material, and its crystal structure is composed of a silicon-oxygen tetrahedron. Because of its excellent properties, low density, large specific surface area, controllable pore structure, large surface permeability, environmental protection, and good fire resistance [154–156], SiO_2 can be considered an excellent supporting material. The three-dimensional network structure of SiO_2 prevents PCM from leaking and flowing during phase transition. Therefore, SiO_2 -based composite phase change materials, like other clay mineral-based composites, have ideal thermal properties and are mainly used for energy-saving and solar energy storage.

Nanometer flower-shaped folded mesoporous SiO_2 (NFMS) has a unique pore structure, large specific surface area, and pore

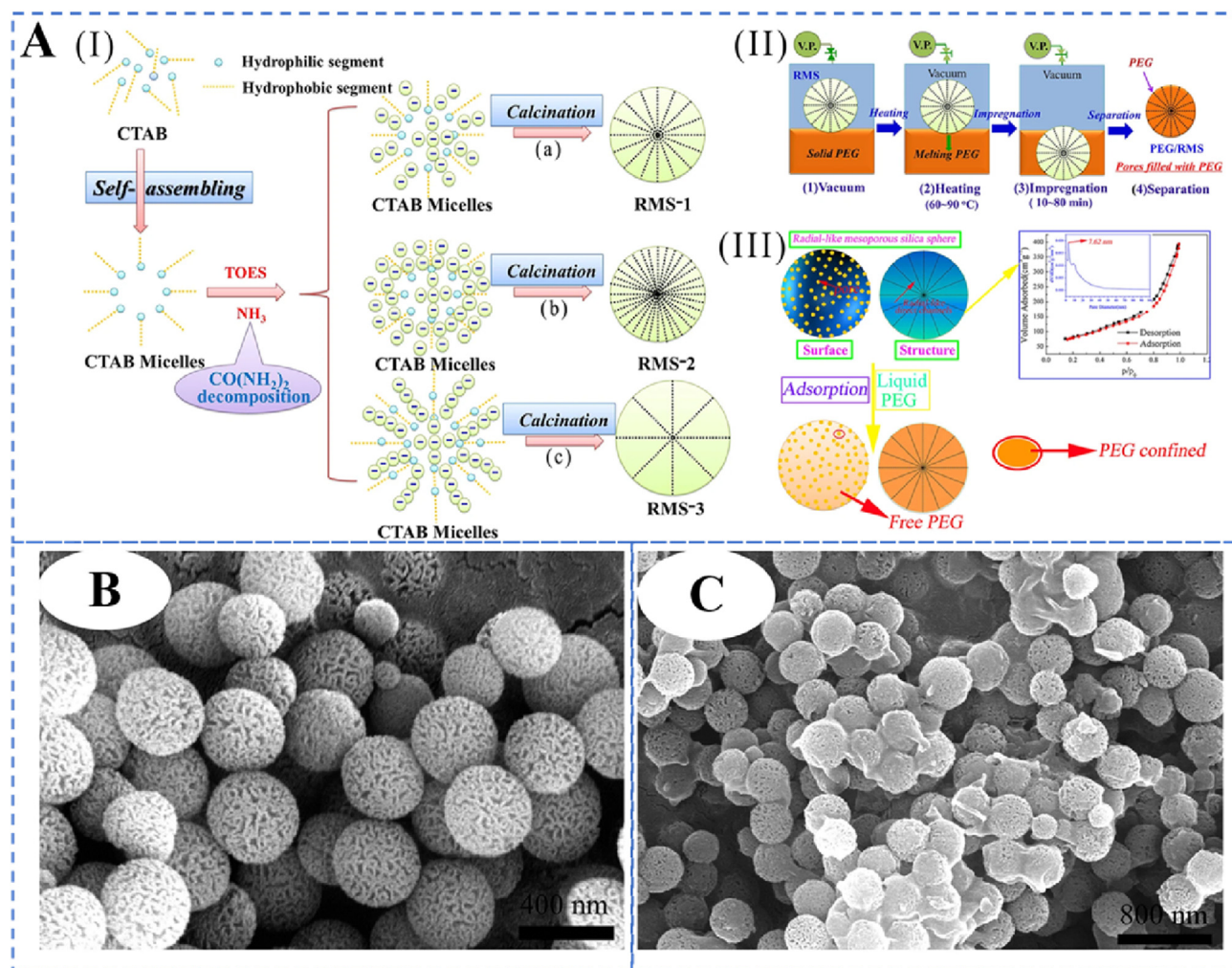


Fig. 20. (I) The schematic formation processes of RMS; (II) the vacuum impregnation treatment for preparing phase change composites (VP: vacuum pump); (III) schematic formation mechanism of the PEG/RMS composite (A). SEM images of RMS spheres before (B) and after adsorption of PEG (C) [158]. Copyright 2015, Springer Nature.

volume is easy to manufacture and low cost. GAO et al. [157] prepared PEG/NFMS composites with stable shapes using NFMS as carrier and polyethylene glycol (PEG) as phase change material. The results show that PEG is wholly absorbed in the internal pores of NFMS through physical adsorption, and NFMS does not affect the crystal structure of PEG. As the DSC test shows, although NFMS has a certain limiting effect on the activity of PEG molecules, the melting enthalpy of PEG/NFMS is as high as 136.6 J/g. In addition, the thermogravimetric analysis showed that no significant mass loss was observed at 20–190 °C, which indicated that the composite had good thermal stability. Min et al. [158] used CTAB surfactant as a template and TEOS as a SiO₂ source to synthesize radial mesoporous SiO₂ (RMS) spheres for support of shape-stable composites (Fig. 20A). A novel ss-PCM consisting of PEG and RMS was prepared by vacuum impregnation (Fig. 20B and C). DSC results show that PEG/RMS ss-PCM has high latent heat, suitable phase transition temperature, good thermal stability, excellent chemical compatibility, and thermal stability. Among them is that SiO₂ nanospheres with a fibrous structure are easier to make molecules reach adsorption sites than other mesoporous materials because of their large radial pore size. It shows that mesoporous SiO₂ is an excellent supporting material.

5.2.2. Effect of MgO-based porous materials on the performance of PCMs

Mesoporous MgO has an abundant pore structure, high specific surface area, and can provide many surface active sites, which is an excellent structure with good adsorption characteristics [159]. Therefore, MgO is an excellent supporting material for the adsorption of PCM.

Liu et al. [160] NaNO₃/MgO composites were prepared by a new low-temperature ceramic sintering process using NaNO₃ as PCM and MgO as supporting material. It is found that the composites have excellent mechanical properties, and the compressive strength is as high as 100 MPa. XRD spectra before and after 500 thermal cycles show no new phases are formed except NaNO₃ and MgO, indicating that NaNO₃ and MgO are chemically compatible at high temperatures. It is noteworthy that the melting enthalpy of the composites remains at 89% of the initial value after 500 thermal cycles, which indicates that NaNO₃/MgO composites have excellent thermal reliability.

In order to enhance the thermal conductivity of composites, Ye et al. [161] prepared Na₂CO₃/MgO shaped PCMs, and used multi-walled carbon nanotubes (MWCNT) as thermal conductivity enhancers. The experimental results show the thermal conductivity of

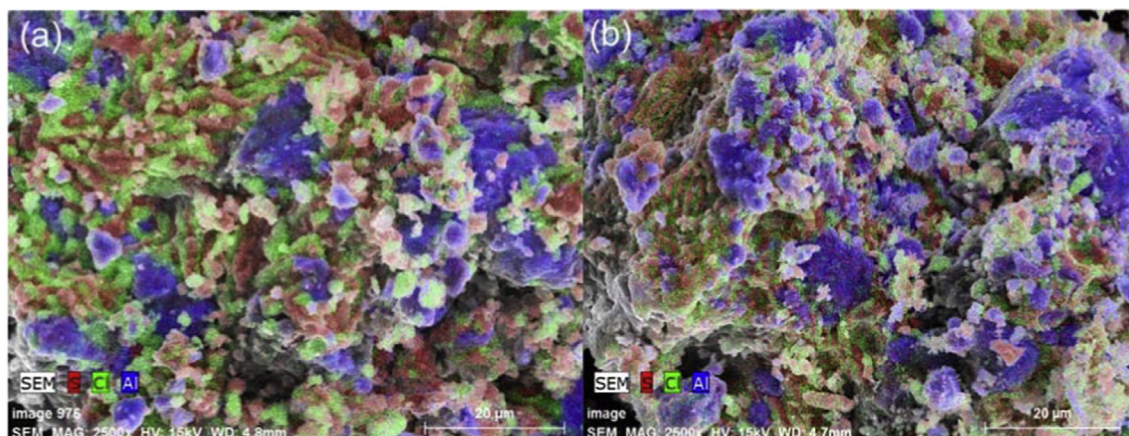


Fig. 21. SEM-EDS analysis of Na₂SO₄-NaCl/α alumina after sintering (a); after 20 thermal cycles (b) [165]. Copyright 2018, Elsevier.

the composites containing 0.5 wt% WMCNT can reach $1.489 \text{ W m}^{-1} \text{ K}^{-1}$ at 120°C . It is worth noting that MgO skeleton material can provide the porous skeleton for attached carbonate and significantly improve the thermal conductivity of composite phase change materials by enhancing the convection coefficient of the porous skeleton and radiation heat transfer in pores. In addition, the composite also has good thermal stability, and its mass loss is less than 5%.

Similarly, Li et al. [162] NaLiCO₃/MgO composite phase change materials were prepared using NaLiCO₃ as PCM, MgO as skeleton support material, and graphite sheet as thermal conductivity enhancer. They used two kinds of MgO (light MgO and heavy MgO). The results show that light MgO can provide greater particle migration and rearrangement due to its higher specific surface area, making the distribution of salt, MgO, and graphite flakes more uniform. Smaller MgO particles produce smaller internal pores and a harder structure, thereby better accommodating PCM and a more stable composite structure during thermal cycling. At the same time, light MgO matrix composites have higher thermal conductivity and mechanical strength than MgO matrix composites. The addition of a graphite sheet increases the thermal conductivity of composites, but the mechanical strength decreases.

5.2.3. Effect of Al₂O₃-based porous materials on the performance of PCMs

Porous Al₂O₃ is a porous material with a large specific surface area, strong adsorption capacity, excellent thermal stability, compression resistance, and thermal conductivity and is an ideal matrix support material [163].

Gong et al. [164] prepared nano-porous Al₂O₃ foam ceramics by calcination method and combined them with expanded graphite to prepare composites with multilevel pore structure. The adsorbed and encapsulated 1-octadecyl alcohol (OD) phase change materials. The results show that OD is mainly adsorbed on the pores, folds, and surfaces of Al₂O₃@EG, and OD and Al₂O₃@EG are physically combined without chemical reaction. The thermal conductivity and thermal diffusivity of Al₂O₃@EG were $1.876 \text{ W m}^{-1} \text{ K}^{-1}$ and $1.17 \text{ mm}^2 \cdot \text{s}^{-1}$, respectively, which were 4.5 and 4.6 times higher than those of pure OD ($0.415 \text{ W m}^{-1} \text{ K}^{-1}$ and $0.25 \text{ mm}^2 \cdot \text{s}^{-1}$), respectively when the addition of Al₂O₃@EG was 10 wt%. After 200 cold and hot cycles, the composite phase change materials still have a high energy storage density. Jiang et al. [165] prepared Na₂SO₄-NaCl/α-Al₂O₃ phase change composites with a high-temperature setting by cold pressing sintering method for a solar thermal power station. The results show that Na₂SO₄-NaCl supported by α-Al₂O₃ has excellent stability, and the weight loss rate of composites is only 0.50% after 20 thermal cycles. It can be seen from the SEM-EDS diagram of the composite phase change material (Fig. 21) that Na₂SO₄-NaCl is still uniformly distributed in the composite material after 20 thermal cycles. The results of DSC and XRD show that the phase separation and chemical reaction of Na₂SO₄-NaCl in the composites do not occur, and the composites have good thermal stability. At the same time, the addition of α-Al₂O₃ can reduce the undercooling of Na₂SO₄-NaCl salt.

5.2.4. Effect of SiC-based porous materials on the performance of PCMs

Porous SiC materials have open channels and high thermal conductivity walls and strong mechanical properties, corrosion

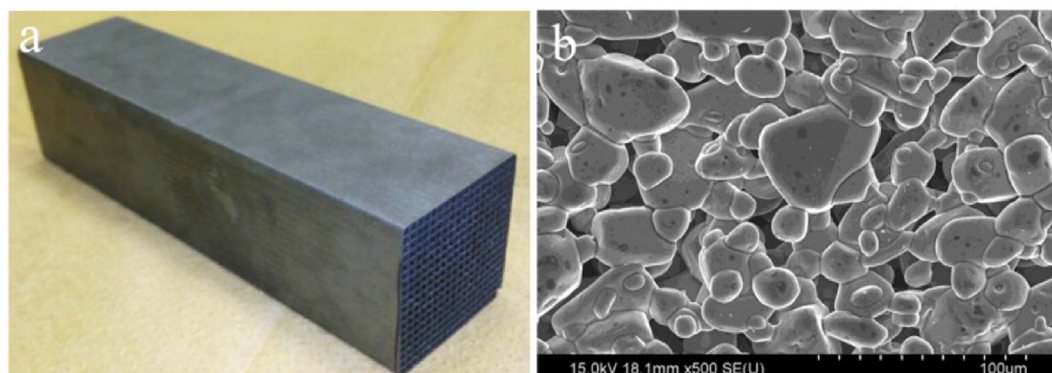


Fig. 22. (a) Photographs of SiC and (b) SEM images of SiC walls [166]. Copyright 2015, Elsevier.

resistance, and oxidation stability. Fig. 22 is an SEM image of porous SiC [166]. SiC can be added to some porous matrix composites by using its excellent thermal properties to form through thermal conductive chains, thus improving the thermal conductivity of the composites. Therefore, SiC material is an excellent skeleton material of composite phase change materials.

Li et al. [166] prepared composite PCMs by vacuum infiltration with nitrate mixture $\text{KNO}_3/\text{NaNO}_3$ as phase change materials and SiC honeycomb ceramics as supporting materials. The experimental results show that the melting and solidification temperatures of the composite phase change materials are slightly different from $\text{KNO}_3/\text{NaNO}_3$, and the latent heat of melting and solidification of the composite phase change materials is 72.8 kJ/kg and 70.3 kJ/kg, respectively. However, the latent heat of composite phase change materials is not high. With the increase of SiC skeleton mass fraction, composite phase change materials' energy storage and release rates increase. The storage and release time of composite phase change materials containing 30 wt% SiC is 52.8% and 58.3% less than pure phase change materials, respectively. Wu et al. [167] used SiC foam ceramics to stabilize the shape of Na_2SO_4 inorganic salt by melt infiltration. The SiC foam ceramics have a three-dimensional porous structure and have good chemical compatibility with Na_2SO_4 phase change materials. In addition, the excellent wettability between SiC foam ceramic carrier and Na_2SO_4 phase change material ensures good physical adhesion between them. Luo et al. [168] embedded PCMs contained SiO_2 nanoparticles in graded ultra-light SiC ceramics, in which $\text{LiNO}_3/\text{NaCl}$ eutectic was used as PCMs. The results show that the thermal conductivity of PCMs ($2.78 \text{ W m}^{-1} \text{ K}^{-1}$) is 159% higher than that of PCMs ($1.073 \text{ W m}^{-1} \text{ K}^{-1}$). The presence of SiO_2 nanoparticles slightly deteriorates the thermal conductivity of the composites. Still, the continuous heat transfer path formed by the porous silicon carbide skeleton significantly enhances the thermal conductivity. They also applied the prepared composite materials to store solar energy. Compared with pure PCMs (25.2%), the composite exhibits a high solar energy absorption rate of 76.3%. Therefore, they believe that the composite material can effectively store heat and solar energy.

5.3. Application of porous ceramic material-based composite PCMs

Ceramic matrix composite PCMs generally have excellent mechanical and thermophysical properties, high energy storage density, and secondary processing ability. Therefore, the new ceramic matrix composite phase change materials will develop the application of energy storage materials in solar energy utilization, new building materials, power peak shaving, industrial waste heat, waste heat recovery, and other fields. According to the types of phase change materials and different phase-change temperatures, composite PCMs can also be used as sensible heat refractory bricks in dense industrial furnaces (hot blast stoves) and heat accumulators in solar power generation systems space stations and temperature-controlled electronic radiators, etc.

5.3.1. Application in the field of construction

It is known that a large amount of energy is consumed in buildings due to heating-cooling and air-conditioning systems [169]. This reveals the necessity of eco-friendly approaches that enable more efficient use of energy and renewable energy sources (solar energy, etc.) in buildings. Using PCMs for thermal energy storage (TES) is one of the effective ways to use solar energy [170,171]. The PCMs absorb heat from the environment during melting and release the absorbed heat during solidification, storing the energy as latent heat in their bodies. This property makes them very attractive for residential structures. Due to their TES capability, using wall elements containing PCM in buildings may be very

effective since about 30% of the heat loss occurs through the walls of the building [152,172]. There are two primary materials used to weave the walls of buildings; one is cement-based bricks, and the second is clay-based sintered bricks. Clay-based sintered bricks significantly disadvantage PCM applications due to firing temperatures exceeding 900°C [173].

Gencel et al. [173] used microfine expanded vermiculite (MEV) impregnated with the lauric acid-myristic acid eutectic mixture (LA-MA) and prepared FSPCM by direct impregnation method. FSPCM with different mass proportions was then used instead of sand to produce cement-based mortar. The results showed that the material's thermal conductivity, bulk density and compressive strength were reduced by 60.93%, 39.2% and 85.94%, respectively, after incorporating 45% of FSPCM. However, in terms of compressive strength, all specimens with composite PCM could meet the standard requirements. Thermal regulation performance tests showed that the produced cementitious mortars containing FSPCMs could provide thermal comfort by reducing indoor temperature fluctuations, thereby reducing the building's thermal and cooling loads and thus saving energy. In their study, PCMs were encapsulated in building materials using attapulgite [174] and fly ash [175] as support materials. The results show that the composite phase change materials have good mechanical and thermal properties. Therefore, they have important potential for thermal regulation and energy saving in buildings.

Xie et al. [176] prepared a new type of EV matrix composite by vacuum impregnation method. The PCM is composed of $\text{Na}_2\text{SO}_4 \cdot \text{H}_2\text{O} - \text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$ binary eutectic salt hydrate, used for building thermal insulation. The results show that the composite PCM board can effectively keep the room temperature comfortable and affect the delay time. Compared with the control room, the room with the PCM board has a 4.8°C peak temperature reduction. The slow heat release rate also shows that the composite PCM has excellent heat storage capacity and excellent insulation capacity. Moreover, the material has the advantages of low undercooling, uniform phase, large heat storage, good insulation performance, suitable phase transition temperature, and non-flammability. Yao et al. [177] Using paraffin wax and expanded perlite as raw materials, composite phase change materials were prepared by horizontal vacuum absorption rotating roller. DSC results show that the latent heat of PCM used in the building envelope is more than 50 J/g, which meets the standard of feasible PCM-based wallboard. The numerical model is established by TRNSYS software, and the experimental results are verified. Numerical analysis shows that the average temperature of a building with a floor area of 4000 m^2 can be reduced by 9.22 K during the running time from 7:00 to 18:00 in two months. In addition, all incremental costs of PCM wallboard can be recovered within 5.84 years. The results show that ss-PCM has the potential to improve building thermal comfort and energy saving. Kang et al. [178] developed a biomass phase change material with fumed silica as an adsorption substrate, which can be applied to building energy-saving concrete. When the temperature is above 300°C , the weight loss of biomass phase change materials with fumed silica is reduced by more than 27%. The thermal conductivity is doubled compared with that without fumed silica. Sari et al. [147] studied a capric acid (CA)-palmitic acid (PA) eutectic mixture/fumed silica composite phase change material, which still showed good thermal stability after 1000 thermal cycles. At the same time, the thermal conductivity of phase change materials is also significantly improved after being modified by fumed silica and carbon nanotubes. Qu et al. [179] prepared composite phase change materials for new phase change foam concrete by using fumed silica to adsorb paraffin wax. Through morphological and leakage tests, it is found that the composite phase change material with paraffin content of 45 wt% has the best

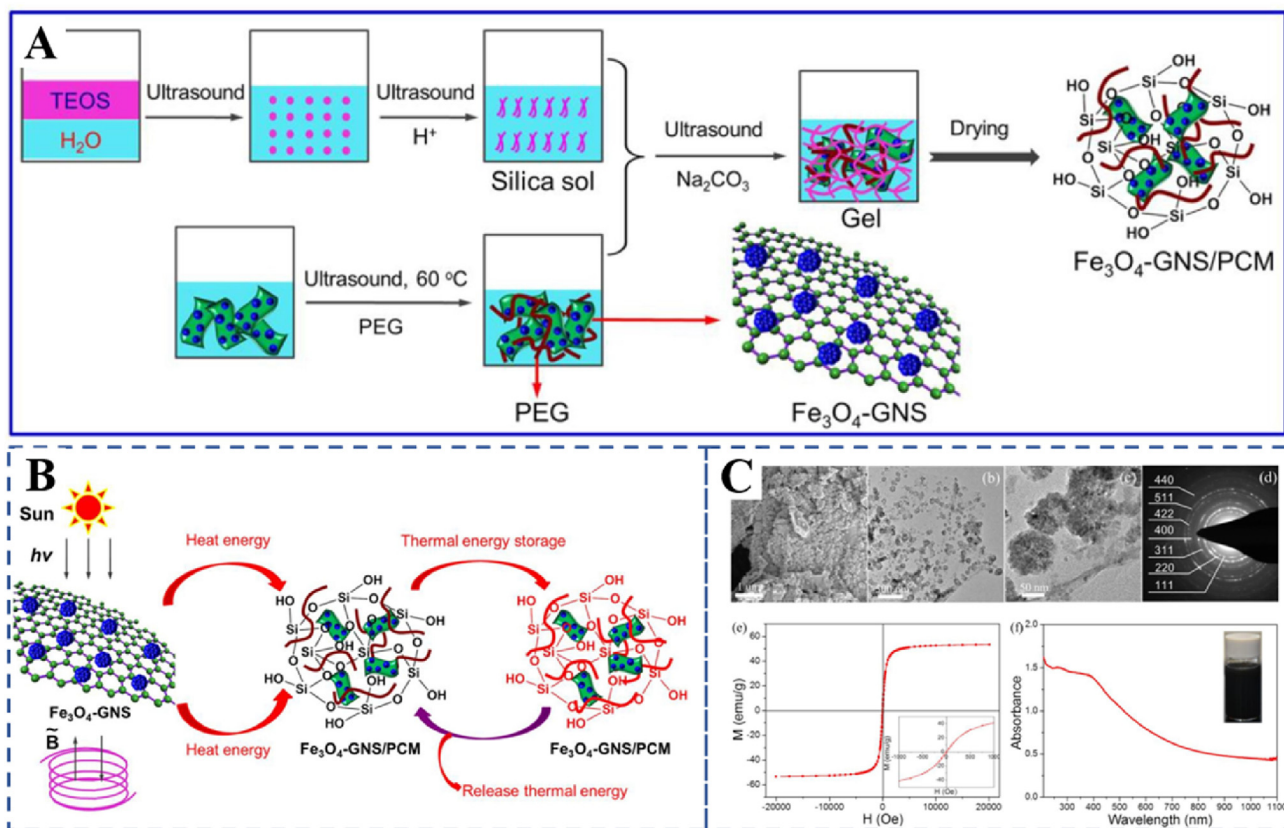


Fig. 23. (a) Synthesis scheme of composite material. (B) Schematic diagram of magnetocaloric and photothermal energy conversion and storage. (C) SEM and TEM of the composites (Above). Magnetization circuit of composites at 300 K and ultraviolet-visible-near infrared absorption spectra of composites in ethanol (Below) [180]. Copyright 2019, Elsevier.

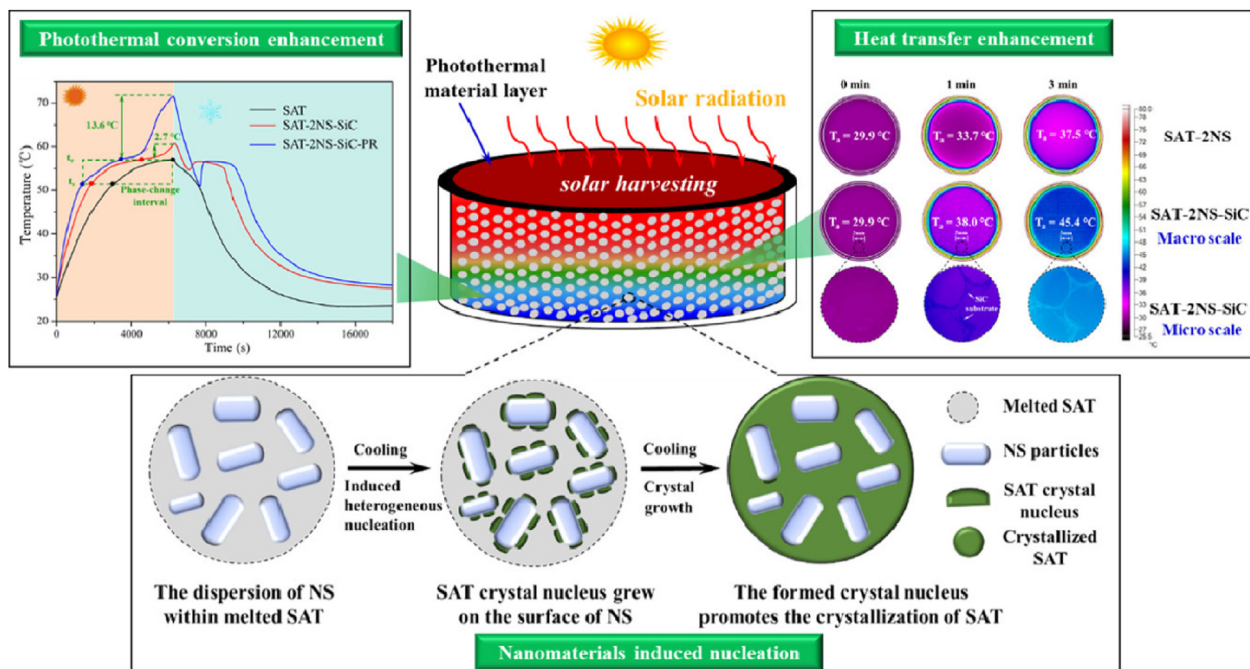


Fig. 24. New stable SAT-based composite PCM with fast solar charging performance [181]. Copyright 2022, Elsevier.

adsorption capacity and condensation performance and can meet the practical requirements of low thermal conductivity and suitable phase change temperature.

5.3.2. Applications in the field of solar energy

Wang et al. [180] prepared multifunctional nanocomposites with PEG/SiO₂ as morphologically stable phase change materials,

Table 6

Summary of work on porous ceramic matrix/PCM composite phase change materials.

Ref.	Support	PCM	T_m of PCM ($^{\circ}\text{C}$)	T_f of PCM ($^{\circ}\text{C}$)	K_{PCM} ($\text{W m}^{-1} \text{K}^{-1}$)	K_{comp} ($\text{W m}^{-1} \text{K}^{-1}$)	H_{PCM} (J/g)	Loading (wt %)	Outcomes
[127]	Diatomite	Paraffin	70	—	—	0.5–0.7	—	—	The composites have good chemical compatibility and can maintain good thermal properties after many thermal cycles.
[128]	Diatomite	KNO_3	334	—	0.45	—	100	65	The adsorption rate of PCM was 65%. After 50 thermal cycles, the weight loss rate is only 4.1%.
[129]	Diatomite	CA-LA	19.09	17.95	—	0.219–0.467	141.5	53.6	The maximum adsorption capacity of PCM was 53.6%. After 200 thermal cycles, it still has good thermal properties and chemical structure.
[130]	EV	CA-PA-SA	19.6	18	—	0.362	158	70	Composite phase change materials have good chemical stability and thermal reliability and can still show enough thermal stability after 200 thermal cycles.
[131]	EV	NaNO_3	306.4	—	—	0.7–3.52	178.6	734.6	The thermal conductivity is improved, and the adsorption capacity is as high as 734.6%.
[132]	EV	LA	41.87	37.88	—	0.28–0.50	197.2	68	The composites have good chemical compatibility and can maintain good thermal properties after 200 thermal cycles.
[133]	EP	PEG	—	—	0.254	0.463–0.584	—	62.53–69.36	When the loading of PCM is over 60%, the composites have excellent thermal stability.
[134]	EP	NaNO_3	307.5	305.9	0.5	0.84	178.4	114.07	The thermal conductivity of the composites is improved, and the composites have good thermal stability.
[135]	EP	Paraffin	62.8	—	—	—	131.7	200.00	After 300 thermal cycles, the phase change temperature and latent heat decreased by 0.96% and 2.3%, respectively.
[136]	EP	RT27	25.1	25	0.166	0.149–0.247	154	60	The thermal conductivity of the composites is improved, and the composites have good thermal stability.
[139]	SiO_2	PEG	63.1	35.7	—	—	202.1	77	The composites have good chemical compatibility, and the adsorption capacity is as high as 76.7%.
[140]	SiO_2	PEG	59.15	40.04	—	—	191.5	79.20	The composites have good chemical compatibility, and the adsorption capacity is as high as 79.2%.
[143]	MgO	NaNO_3	306.8	303.76	—	—	168.05	62	After 500 thermal cycles, it still shows enough thermal stability, and the compressive strength is as high as 100 MPa.
[131]	MgO	Na_2CO_3	—	—	—	0.881–1.489	—	60	Improve thermal conductivity and good thermal stability.
[144]	MgO	NaLiCO_3	498.8–512.4	—	—	2.60–5.81	347.9	50	Lightweight MgO can better accommodate PCM during thermal cycling, and the thermal conductivity of composites is improved by adding graphite sheets.
[146]	Al_2O_3	OD	58.54	52.18	0.415	1.876	238.51	90	The thermal conductivity of the composites is increased by 4.5 times, and the composites still have a high energy storage density after 200 thermal cycles.
[147]	Al_2O_3	$\text{Na}_2\text{SO}_4\text{--NaCl}$	625.3	614.7	—	—	263.3	50	After 20 thermal cycles, PCM can still show enough thermal stability and reduce the undercooling degree of PCM.
[148]	SiC	$\text{KNO}_3/\text{NaNO}_3$	222.6	220.4	—	—	106.3	70	Energy storage and energy release are saved by 52.8% and 58.3%, respectively.
[149]	SiC	Na_2SO_4	888.3	—	—	5.5	161	—	The composites have good chemical compatibility.
[150]	SiC	$\text{LiNO}_3/\text{NaCl}$	225	—	1.073	2.78	357.6	95	The thermal conductivity increases by 159%, and the solar energy absorption rate improves.

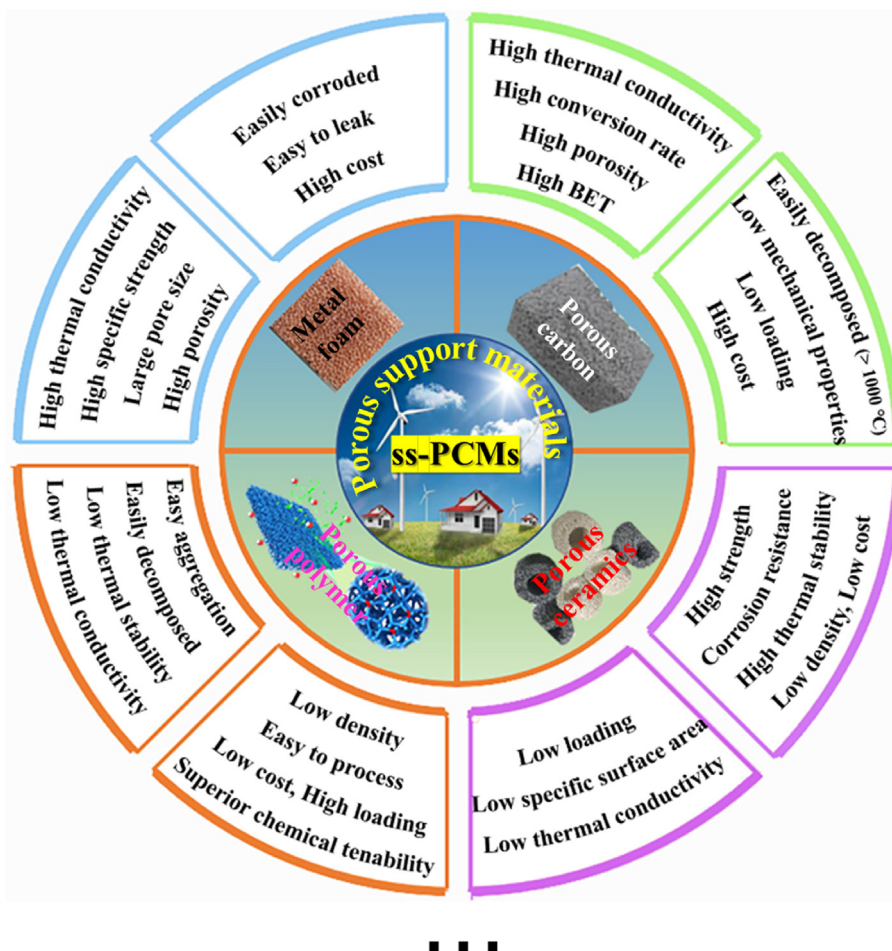


Fig. 25. Summary of advantages and disadvantages of porous materials as support materials for SS-PCMs.

and Fe_3O_4 functionalized graphene nanosheets (Fe_3O_4 -GNS) as energy converters. The preparation process and magnetocaloric and photothermal energy conversion and storage principles are shown in Fig. 23A and B. They studied the properties of the composites, as shown in Fig. 23C. The results show that the magnetocaloric effect of Fe_3O_4 nanoparticles and the light capture characteristics of graphene in Fe_3O_4 -GNS make it realize efficient electromagnetic and solar energy conversion and storage under alternating magnetic fields or sunlight irradiation. Due to the melting phase transition of PEG, the heat generated during the energy conversion process is stored in the phase change materials with stable morphology. This novel energy conversion and storage system will have a good application prospect in energy management, such as electromagnetic or solar energy driving equipment. The application currently stays in the small-scale experimental stage, so further efforts are needed for large-scale applications.

In order to improve the photothermal conversion efficiency and heat transfer efficiency of solar cells, to improve the solar energy storage efficiency. Cui et al. [181] prepared SAT-NS-SiC-PR composites by using nano-SiO₂ (NS) as a nucleating agent, SiC foam as skeleton material and thermal conductivity enhancer, and carbon nanotubes as a photothermal enhancer (Fig. 24). The results show that SAT undercooling decreases to 1.4 °C, and thermal conductivity increases by 140.6%. The charging efficiency of solar energy is increased from 59.3% to 75.4%, the recovery period of solar energy storage using composite materials is only 3.1 years, and the carbon dioxide emission can be reduced by about 1379.1 kg CO₂-eq/year.

5.3.3. Applications in its field

Ceramic matrix composite PCMs are widely used for building energy saving. Some other applications are also studied, such as waste heat recovery in industry and application in diving suits. Zhao et al. [182] coupled FA-based FSPCM with shell-and-tube heat exchangers to recover waste heat in industry. Good results of the coupling system are obtained in the research. Wang et al. [183] studied the effect of EP-based FSPCM on diving suits. The results show that the prepared fatty acid/EP FSPCM can meet the temperature requirements of diving suits.

5.4. Summary

Three primary skeleton materials are used to prepare ceramic-based composite PCMs: clay mineral ceramic materials (diatomite, expanded vermiculite, and expanded perlite); Oxide ceramic materials (SiO₂, MgO, Al₂O₃); Non-oxide ceramic material SiC. Choosing different types of matrix materials determines the different properties of composite phase change materials.

Clay mineral ceramics have a rich porous structure, high-temperature resistance, and good wettability with phase change materials. However, the thermal conductivity of such matrix materials is low and usually lower than 0.2 W m⁻¹ K⁻¹. This significantly limits the application of this composite phase change material, especially for those applications with high power density requirements. At present, the solution to enhance the heat transfer performance of the composite PCMs is usually to add high thermal

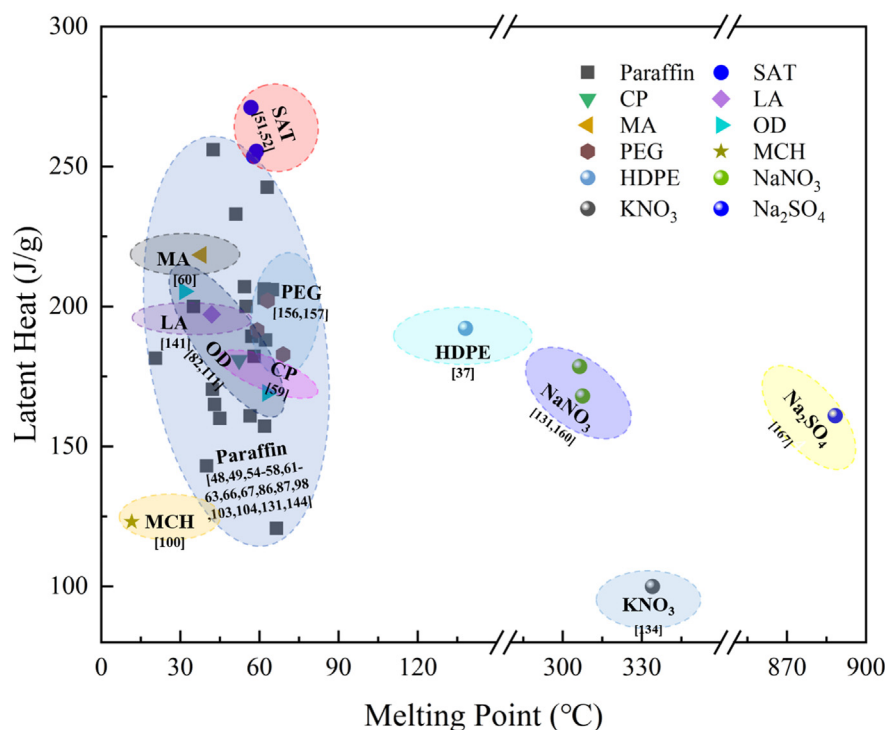


Fig. 26. Different melting points and latent heat ranges of PCMs used by different researchers.

conductivity particles, such as graphite, silicon carbide, etc. More effective measures should be taken to improve the thermal conductivity of shaped phase change materials based on clay minerals.

Ceramics such as SiO₂, MgO, Al₂O₃, and SiC can also be used as matrix materials by virtue of their newly formed porous skeletons. The ability of the matrix to adsorb phase change materials is weak, which is limited by the porosity of the freshly formed porous matrix. In future research, more attention should be paid to preparing porous supports with high apparent porosity and proper pore size from such matrix materials. Table 6 summarizes the thermal properties of porous ceramic matrix composites.

6. Conclusions and prospects

In recent years, phase change TES has played an essential role in solving the problems of energy shortage and environmental pollution because of its low cost, slight temperature difference, and long-term storage of a large amount of heat energy. However, the undesired characteristics of solid-liquid PCMs during phase transition, including phase separation, undercooling, corrosion, volume expansion, low thermal conductivity, and leakage during phase transition, require additional modification. The encapsulation of PCMs is an effective way to overcome these defects and improve the thermophysical properties. Among them, porous packaging technology can effectively solve the leakage problem of solid-liquid PCMs and further improve the energy storage density and thermal performance, which is a widely accepted and mature technology. Generally speaking, the energy storage and utilization efficiency of porous ss-PCMs are greatly affected by pore size, pore structure, and pore surface properties. Fig. 25 summarizes porous materials' advantages, disadvantages, and applications as ss-PCMs support materials. An absorbent shape-stabilized composite comprising a metal foam, a porous polymer, a porous carbon matrix, and a porous ceramic. This review focuses on the effects of different porous materials on thermal conductivity, latent heat phase transition temperature undercooling shape stability, thermal cycling

stability, and chemical compatibility of PCMs. In addition, the applications of different porous material-based composite phase change materials in various industries are summarized. Although the research of porous composite phase change materials has made some remarkable progress, the problems of low thermal conductivity, long doping time, and low loading of PCMs still need to be solved. Therefore, there are still several problems to be solved in future research.

- (1) At present, the ss-PCM phase transition point coverage is very narrow. As shown in Fig. 26, the phase transition temperature of PCMs used by most researchers is in the low-temperature PCMs from 0 °C to 70 °C and is mainly concentrated on organic PCMs. If ss-PCMs are expected to be widely used in the future, researchers should broaden the selection range of PCMs. Salts, metals, and metal oxides are suitable to heat storage materials at high temperatures.
- (2) Most current studies still stop testing the thermal properties of the initial samples of porous composite phase change materials and then evaluating this new ss-PCMs. This method has disadvantages such as unpredictability, high cost, and long cycle time. A complex two-phase interface exists for the solid-liquid phase change heat transfer process within porous media. Researchers are recommended to use a combination of experimental and MD simulations to explore the complex phase change heat transfer process within porous media and provide theoretical guidance for heat transfer enhancement of phase change energy storage to achieve the thermal demand-oriented material design.
- (3) Pay attention to developing composite phase change materials with low cost, simple process, high energy storage density, suitable phase change temperature, and green sustainability. In the preparation process, the materials' overall properties should be considered suitable for practical application. To better design ss-PCMs composites with thermal storage performance, it is necessary to establish a database of

PCMs composites supported by porous materials. However, there is still a long way to achieve this goal.

- (4) The yield of ss-PCMs is meager, which mainly limits the commercial application of energy conversion. At present, composite shape-setting phase change materials have been developed, which can be used for photothermal, electro-thermal, magnetocaloric, and mixed-mode conversion. Such composites can achieve good shape stability and improve energy conversion efficiency. However, there is little research on these multifunctional energy conversion phase change materials on how to expand their new functions and accumulate data for future commercial production and practical application. It will be an essential research topic in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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